

Teaching Notes 2A

Portfolio Choice and Diversification

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Overview

- Introduction to Risk
- Economics of Portfolio Choice
 - Constructing Portfolios
 - The Capital Allocation Line (CAL)
- The Role of Diversification
 - Systematic and Idiosyncratic
- Portfolio Theory
- Appendix (including “Review of Probability and Statistics”)

Objectives

- Understand how to capture risk
- Develop a portfolio choice intuition that tells us how to choose a portfolio of assets when we are given a set of financial securities with a certain level of risk and return.

Introduction to Risk

How do we measure “Risk”?

- Consider an investor who has \$50,000 to invest. She has the choice of investing the \$50,000 in a risk-free investment or a risky portfolio. The risky portfolio will either halve or double, with *equal* probability.
- The possible payoffs for the two different investments are:

Riskless:

\$50,000 → \$51,500

Risky:

\$50,000 → \$100,000
\$50,000 → \$25,000

- Both the **expected** and **realized return** on the risk-free investment is:

$$(\$51,500 - \$50,000) / \$50,000 = \mathbf{3\%}$$

How do we measure “Risk”?

- The **realized return** on the risky investment is either
 - $r = (\$100,000 - \$50,000) / \$50,000 = 1$ (a gain of 100%); or
 - $r = (\$25,000 - \$50,000) / \$50,000 = -0.50$ (a loss of 50%).
- The **expected return** on the risky investment is (in \$000):

$$E(r) = \frac{1}{2} (100/50 - 1) + \frac{1}{2} (25/50 - 1) = \mathbf{25\%}$$

- **Question:** How do we decide which investment to take?
 - The risky portfolio is expected to earn 22% in excess of the risk-free asset, but it is risky.
 - **How much riskier?** We need a **measure of risk**. The measure of risk we will use for now is the variance or standard deviation.

$$\sigma_R^2 = \text{Var}[R] = p_1 (r_1 - \mu_R)^2 + p_2 (r_2 - \mu_R)^2$$

$$\sigma_R^2 = \text{Var}[R] = \frac{1}{2} (1.00 - 0.25)^2 + \frac{1}{2} (-0.50 - 0.25)^2 = \mathbf{0.56} \quad 6$$

How do we measure “Risk”?

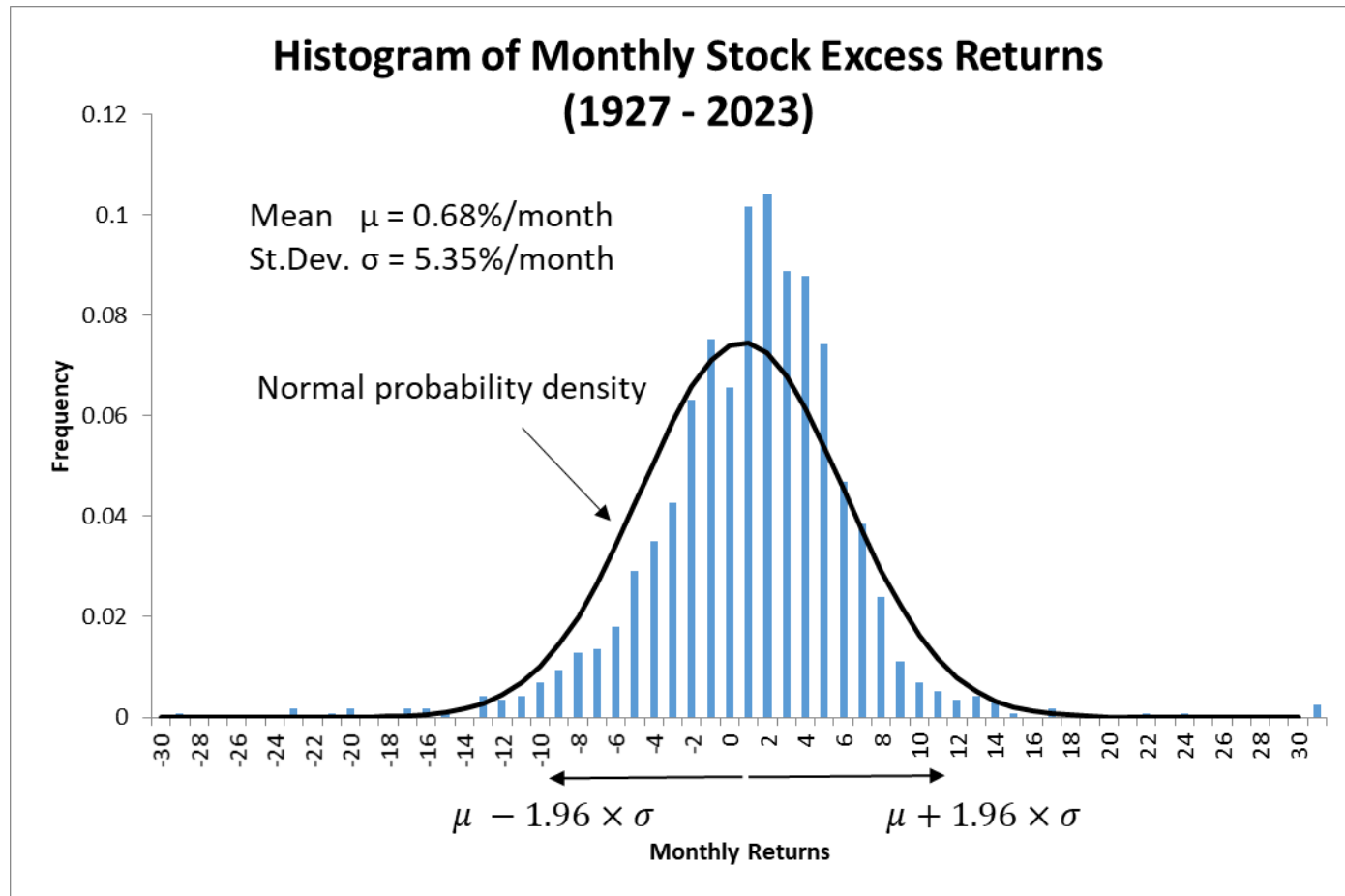
- The standard deviation of the risky asset is the positive square-root of σ_R^2 :

$$\sigma_R = \sqrt{\sigma_R^2} = \sqrt{0.56} = 0.75 = 75\%$$

- **Question:** What is the variance/standard deviation of the risk-free asset?

How do we measure “Risk”?

- For instance, using a normal approximation for the distribution of stock returns, with 95% probability stock returns are between minus/plus 1.96 standard deviations from the mean.



Economics of Portfolio Choice

Portfolio Choice

One Risky Asset and A Risk-Free Asset

- The investor can also construct portfolios of the risky and risk-free asset.
 - Let's calculate the return on a portfolio p consisting of one risky asset and a risk-free asset.
- **Example (Cont'd.):** From our previous example we have:
 - r_A = Return on Risky Asset A (random!)
 - $E(r_A)$ = Expected Return Risky Asset A = 25%
 - σ_A = Standard Deviation = 0.75
 - r_f = Risk-free rate = 3%
 - x = Fraction of portfolio p invested in Asset A
 - $(1 - x)$ = Fraction of portfolio p invested in Risk-free Asset.
- **Note:** x and $(1 - x)$ are the portfolio weights and must sum to 1 (or 100%)!

Portfolio Choice (cont.)

One Risky Asset and A Risk-Free Asset

- The expected return on the risky security and the risk-free asset with weights of 50% in each asset is:

$$E(r_p) = x \times E(r_A) + (1-x) \times r_f$$

$$E(r_p) = \frac{1}{2} \times 0.25 + \frac{1}{2} \times 0.03 = 0.14 = \mathbf{14\%}$$

- This can also be written as:

$$E(r_p) = r_f + x \times (E(r_A) - r_f) \quad (1)$$

$$E(r_p) = 0.03 + \frac{1}{2} \times (0.25 - 0.03) = \mathbf{14\%}$$

- The risk of the combined portfolio is given by:

$$\sigma_p^2 = \text{Var}[P] = V(x \times r_A + (1-x) \times r_f)$$

$$\sigma_p^2 = x^2 \times V(r_A) + (1-x)^2 \times V(r_f) + 2x(1-x) \text{cov}(r_A, r_f) = x^2 \times V(r_A) \quad [\text{Why?}]$$

$$\sigma_p^2 = x^2 \times \sigma_A^2 = 0.5^2 \times 0.75^2 = \mathbf{0.1406} \rightarrow \sigma_p = x \times \sigma_A = 0.5 \times 0.75 = \mathbf{0.375}$$

From Portfolio Choice to.....

The Capital Allocation Line (CAL)

- To get a relation between the expected return and the standard deviation of the portfolio, you can substitute:

$$\sigma_p = x \times \sigma_A \rightarrow x = \sigma_p / \sigma_A$$

into equation (1), giving the **Capital Allocation Line**:

$$E[r_p] = r_f + \frac{\sigma_p}{\sigma_A} \times [E(r_A) - r_f] = r_f + \left[\frac{E(r_A) - r_f}{\sigma_A} \right] \times \sigma_p$$

- For our example the CAL is:

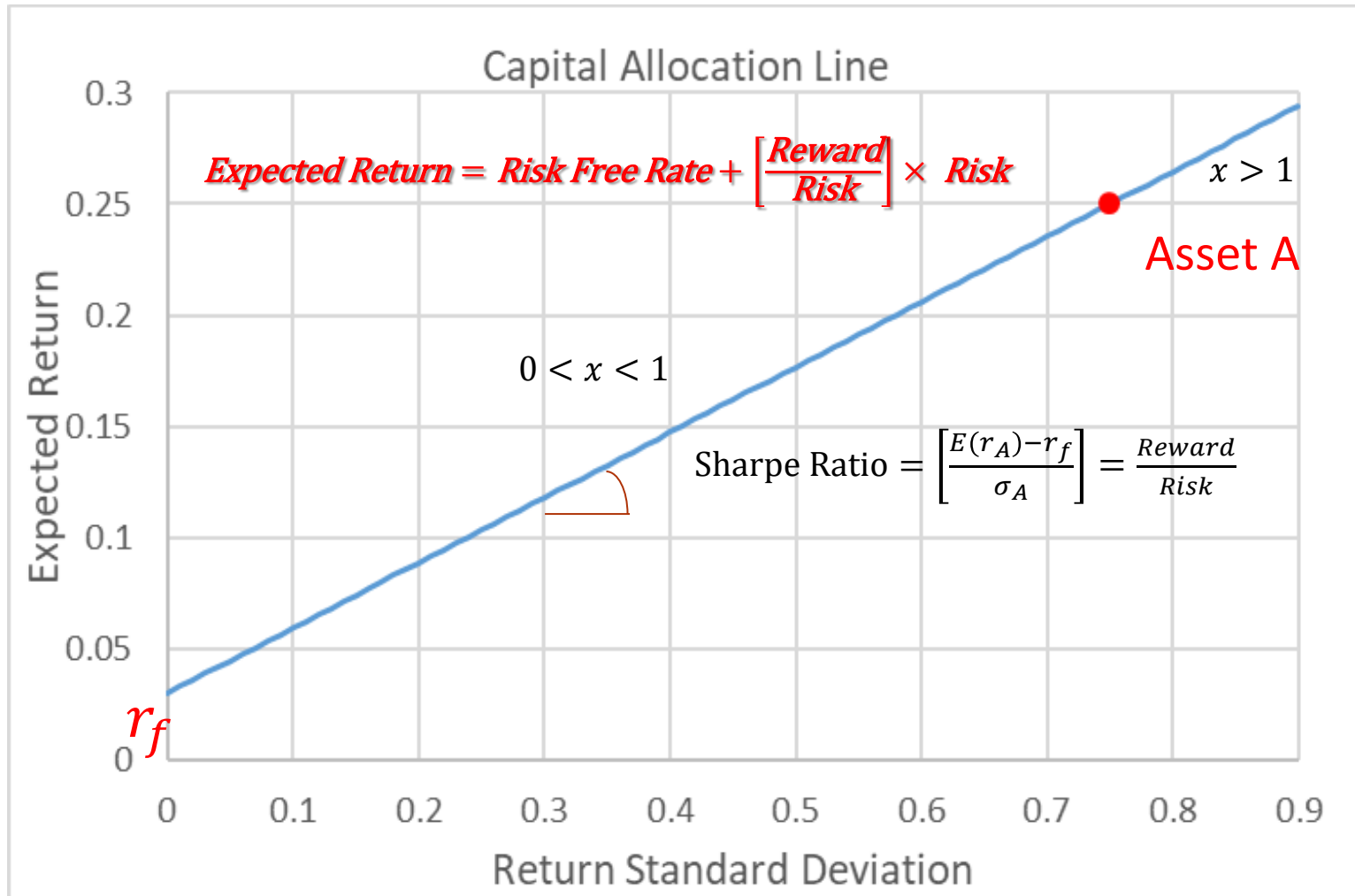
$$E[r_p] = 0.03 + \left[\frac{0.22}{0.75} \right] \times \sigma_p$$

- The intuition for this equation is:

$$\textit{Expected Return} = \textit{Risk Free Rate} + \left[\frac{\textit{Reward}}{\textit{Risk}} \right] \times \textit{Risk}$$

The Capital Allocation Line (CAL) (cont.)

- The CAL shows **all** possible risk-return combinations from a portfolio of one risky-asset and the risk-free asset.



The Capital Allocation Line (CAL) (cont.)

- The **reward-to-risk ratio** is the return premium per unit of risk (standard deviation).
 - The standard term for this ratio is the **Sharpe Ratio**.
- In our example, the Sharpe Ratio of asset A is $0.22/0.75 = 0.293$
 - For a 1% increase in the standard deviation of the portfolio, the expected return of the portfolio increases 0.293%.
- **Question:** What is the Sharpe Ratio on the S&P 500 index?
 - The Sharpe Ratio of the S&P 500 is about 0.4 (= 8%/20%).
 - So you can obtain a 0.4% expected return premium per 1% of standard deviation.

The Role of Diversification

Portfolio Choice

Extending to Two Risky Assets for “Complete” Picture

- The weights with two assets are now given by x_A and x_B .
 - The weights are the fractions of the value of the portfolio p that are invested in Asset A and B, respectively (*i.e.* $x_B = 1 - x_A$).

- The **expected return** on portfolio p is:

$$E(r_p) = x_A \times E(r_A) + x_B \times E(r_B)$$

- The **variance** of the portfolio is:

$$V(r_p) = \sigma_p^2 = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \text{Cov}(r_A, r_B)$$

- **Key:** The **covariance** describes the degree to which two random variables move in the same direction.
 - The risk of a portfolio depends on the covariance between the securities in the portfolio.

Portfolio Choice (cont.)

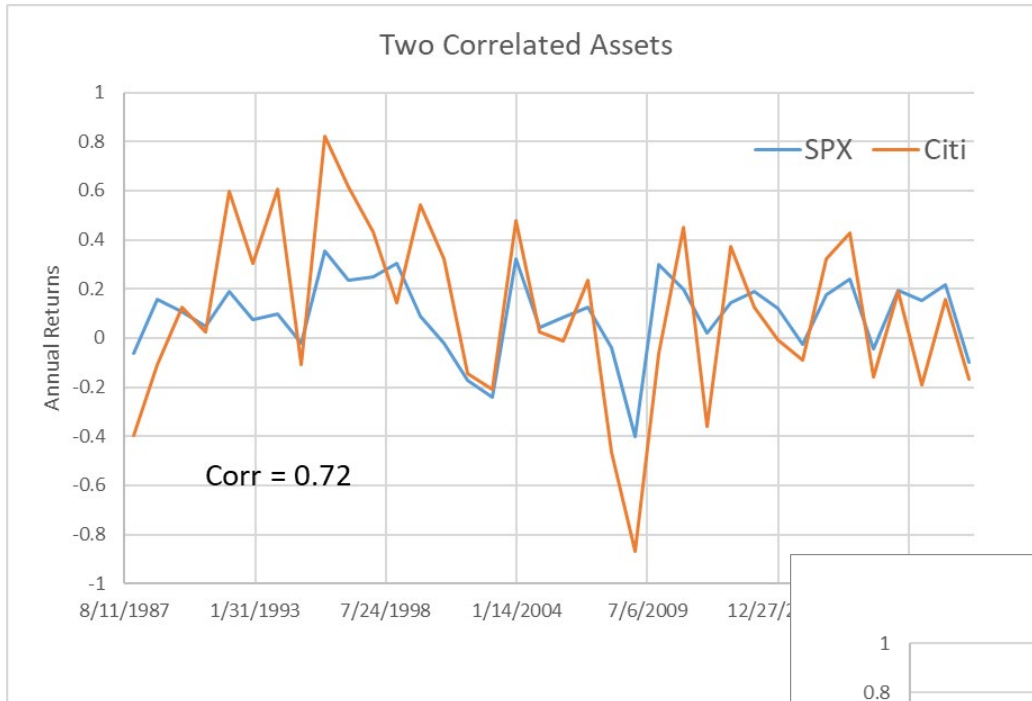
Two Risky Assets

Covariance and Correlation

- If r_A and r_B tend to move **together**, $\text{Cov}[r_A, r_B]$ will be **positive**.
- If r_A and r_B tend to move **opposite**, $\text{Cov}[r_A, r_B]$ will be **negative**.
- **Correlation** is a normalized measure of covariance.
 - It is sometimes easier to think about correlation since it takes values in the interval $[-1,1]$.
 - Random variables that have a correlation of:
 - 1 are perfectly correlated.
 - -1 are perfectly negatively correlated.
 - 0 are uncorrelated.

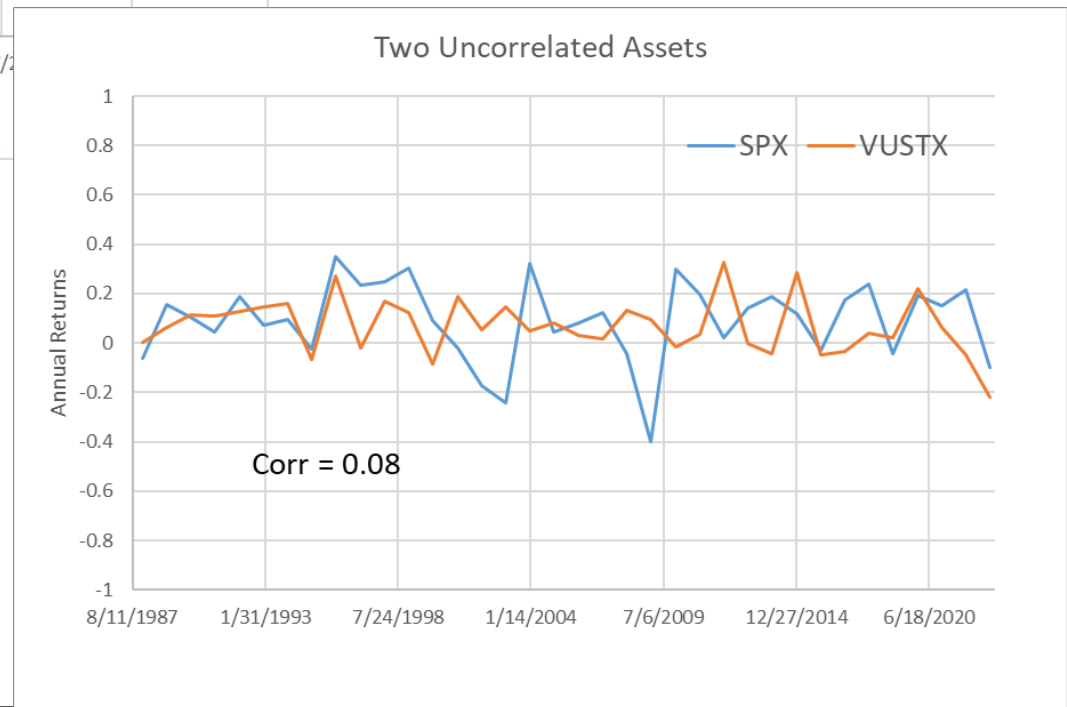
$$\text{Corr}(r_A, r_B) = \rho_{A,B} = \frac{\text{Cov}(r_A, r_B)}{\sigma_A \times \sigma_B}$$

Correlated and Uncorrelated Assets



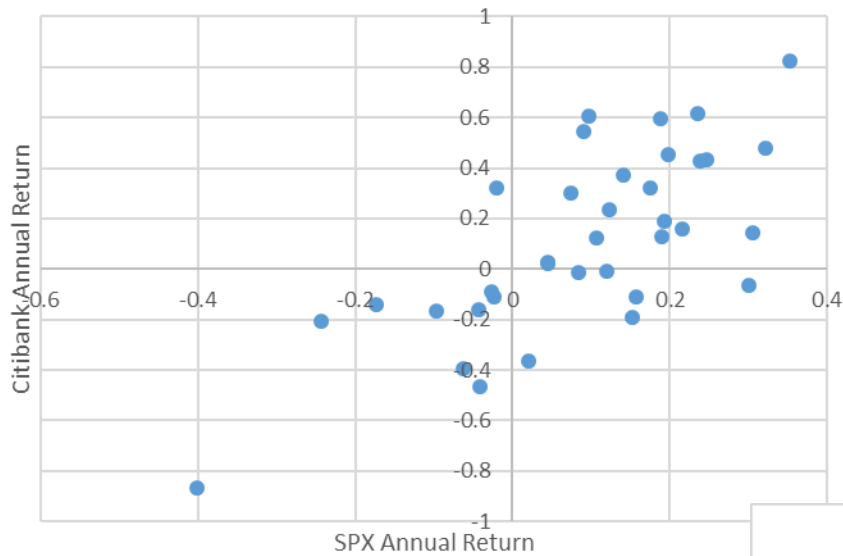
← Citibank and the market index SPX tend to go up and down together

Vanguard long-term Treasury fund (VUSTX) and SPX tend to go up and down quite independently of each other →



Correlated and Uncorrelated Assets

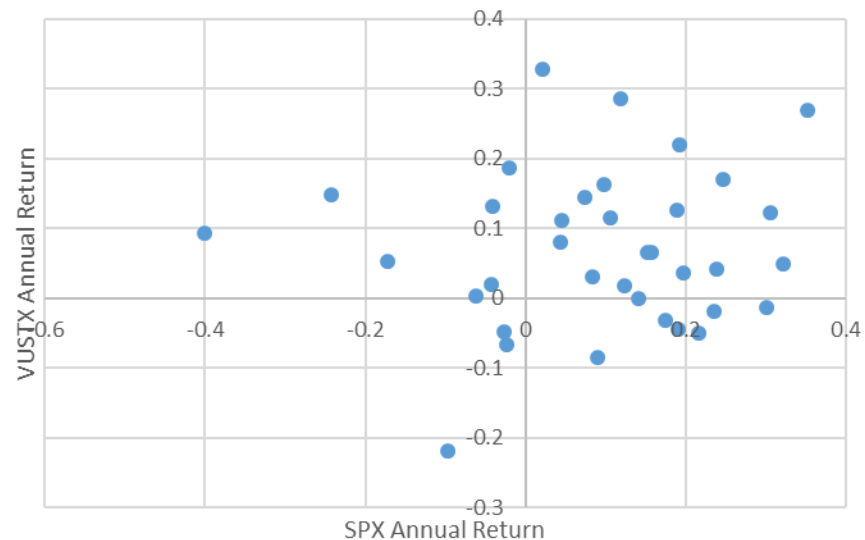
Scatterplot: Citibank (y) vs SPX (x)



Highly correlated returns (Citibank and SPX) tend to “line up” on a scatter plot

Uncorrelated returns (VUSTX and SPX) tend to form a cloud on a scatter plot

Scatterplot: VUSTX (y) vs SPX (x)



Portfolio Choice (cont.)

Two Risky Assets

- **Example:** You invest \$30K in Microsoft (MSFT) stock and \$50K in Johnson and Johnson (JNJ) stock. Suppose you expect the *monthly* return to be 1.2% for MSFT and 0.5% for JNJ.
- The **expected monthly return** on the portfolio is:

$$E(r_p) = x_A \times E(r_A) + x_B \times E(r_B)$$

$$E(r_p) = 30/80 \times 0.012 + 50/80 \times 0.005 = 0.007625 = \mathbf{0.7625\%}$$

- Suppose the standard deviation of monthly returns is 11.6% for MSFT and 6.5% for JNJ, and that the correlation of returns on the two stocks is 0.18. What is the standard deviation of our portfolio?

$$V(r_p) = \sigma_p^2 = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \sigma_A \sigma_B \rho_{A,B} = 0.004179$$

Portfolio Choice (cont.)

Two Risky Assets

- The **standard deviation of the portfolio** is $0.004179^{1/2} = 0.0646 = 6.46\%$ per month which is **less** than either of the two stocks' standard deviations! (How is this possible?)
- The realized returns for August 2004 were in fact -3.9% for MSFT and $+5.6\%$ for JNJ.
 - The **realized return** for the portfolio is:

$$r_p = x_A \times r_A + x_B \times r_B$$

$$r_p = 30/80 \times (-0.039) + 50/80 \times (0.056) = 0.020375 = \mathbf{2.0375\%}$$

Systematic vs. Idiosyncratic Risk

- **Systematic Risk:**

- Returns on different stocks tend to move together due to economy-wide risks.
- This risk is **not diversifiable**.
- Systematic risk = **market risk** or **undiversifiable risk**.

- **Idiosyncratic Risk:**

- Differences in returns which are due to firm-specific risks.
- This risk is **diversifiable**.
- Idiosyncratic risk is also known as **non-systematic risk**, **diversifiable risk**, **firm-specific risk** or **unique risk**.

How Does Diversification Reduce Risk?

- Let's look at the standard deviation for selected foreign stocks and market indexes for the last 15 years

Stock	Standard Deviation	Country	Standard Deviation
ROYAL BANK	17.29%	Canada	20%
BARRICK GOLD	37.48%		
BRITISH PETROLEUM	24.35%	UK	17%
UNILEVER	18.60%		
DEUTSCHE BANK	36.81%	Germany	22%
ENEL	21.46%	Italy	23%
FERRARI	25.72%		
HEINEKEN	19.60%	Netherlands	21%
ASML	38.57%		
LVMH	25.78%	France	21%
TOTAL ENERGY	20.66%		
NESTLE	13.39%	Switzerland	15%
ROCHE	17.98%		
NOKIA	40.10%	Finland	26%
KONE	21.90%		
TOYOTA	24.23%	Japan	16%
SONY	35.65%		

- Standard deviations of individual stocks are normally **more volatile** than the market indexes in which they trade.

How Does Diversification Reduce Risk? (cont.)

- The market portfolio is made up of individual stocks, and as a result its variability does not reflect the average variability of its components
 - Diversification **reduces** variability.
 - It works because prices of different stocks do **not** move exactly together.
- The risk that potentially can be eliminated by diversification is called unique or **firm-specific risk**.
- There is risk that you cannot avoid regardless of how much you diversify. It is called **market risk**.
 - There are economy wide issues that threaten all businesses.
 - Investors are exposed to this risk, no matter how many stocks they hold.

How Does Diversification Reduce Risk? (cont.)

- This plot shows how the standard deviation of a portfolio of average NYSE stocks changes as we change the number of assets in the portfolio.

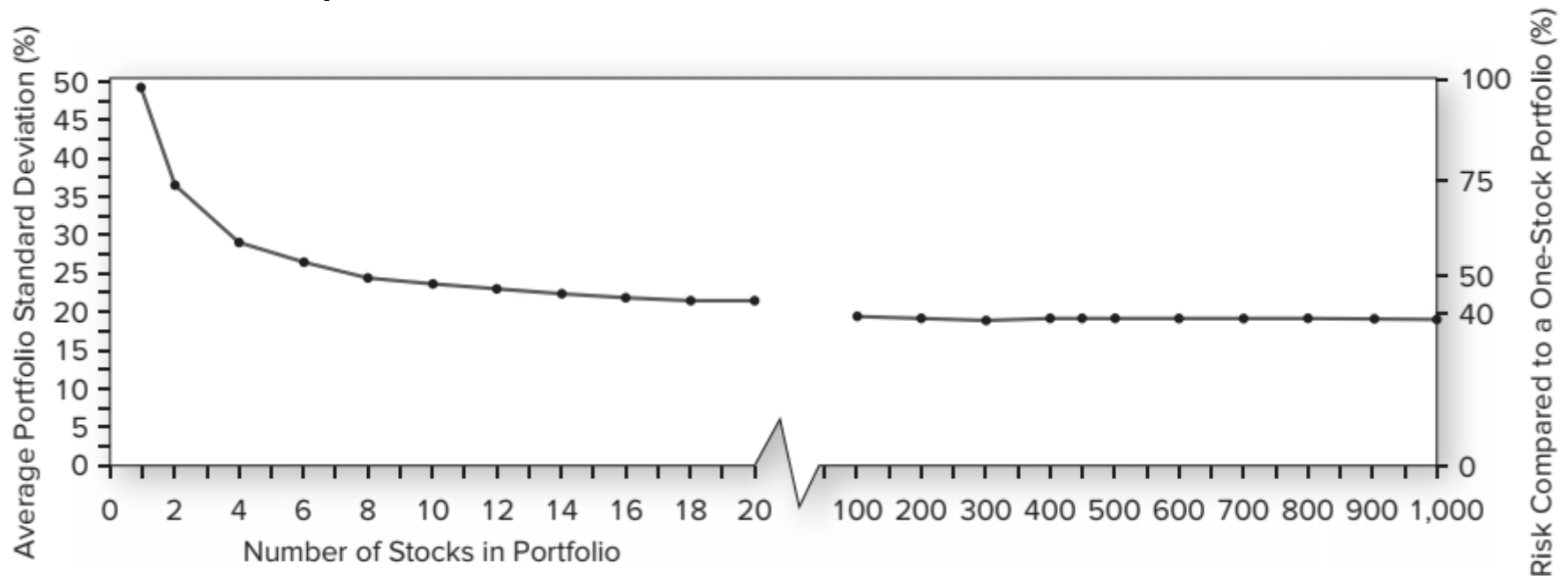


Figure 7.2 Portfolio diversification. The average standard deviation of returns of portfolios composed of only one stock was 49.2%. The average portfolio risk fell rapidly as the number of stocks included in the portfolio increased. In the limit, portfolio risk could be reduced to only 19.2%.

Source: Meir Statman, "How Many Stocks Make a Diversified Portfolio?" *Journal of Financial and Quantitative Analysis* 22 (September 1987).

(Source: Bodie, Kane, and Marcus "Investments" 11 Ed, McGraw – Hill publisher)

How Does Diversification Reduce Risk? (cont.)

- The component of risk that can be diversified away is called the **diversifiable, specific, idiosyncratic** or **non-systematic** risk.
 - The average (annual) standard deviation is 35%.
 - The average (annual) correlation between stocks is 0.35.
 - The average (annual) covariance between stocks is 0.037.
- As number of securities in the portfolio increases, the limiting portfolio standard deviation is $0.037^{1/2} = 0.1924 = 19\%$.
- **Bottom Line:** Since the average covariance is positive, even a very large portfolio of stocks will be risky.
 - The risk that cannot be diversified away is the **systematic risk**.

Portfolio Theory

Intuition: Back to Two Risky Assets Portfolio

- Suppose there are only two assets in the economy. Based on historical data, you estimate statistics on returns:
 - Asset 1: $\mu_1 = 0.10, \sigma_1 = 0.25$
 - Asset 2: $\mu_2 = 0.15, \sigma_2 = 0.30$
 - Correlation: $\rho_{12} = 0.25$
- Identify the combinations of portfolio expected return and standard deviation that you can obtain by combining these assets.
- The set of portfolios you can choose have as their random return: $r_p = x \times r_1 + (1-x) \times r_2$
- For a given portfolio choice (*i.e.*, choice of “x”), the portfolio’s expected return is:

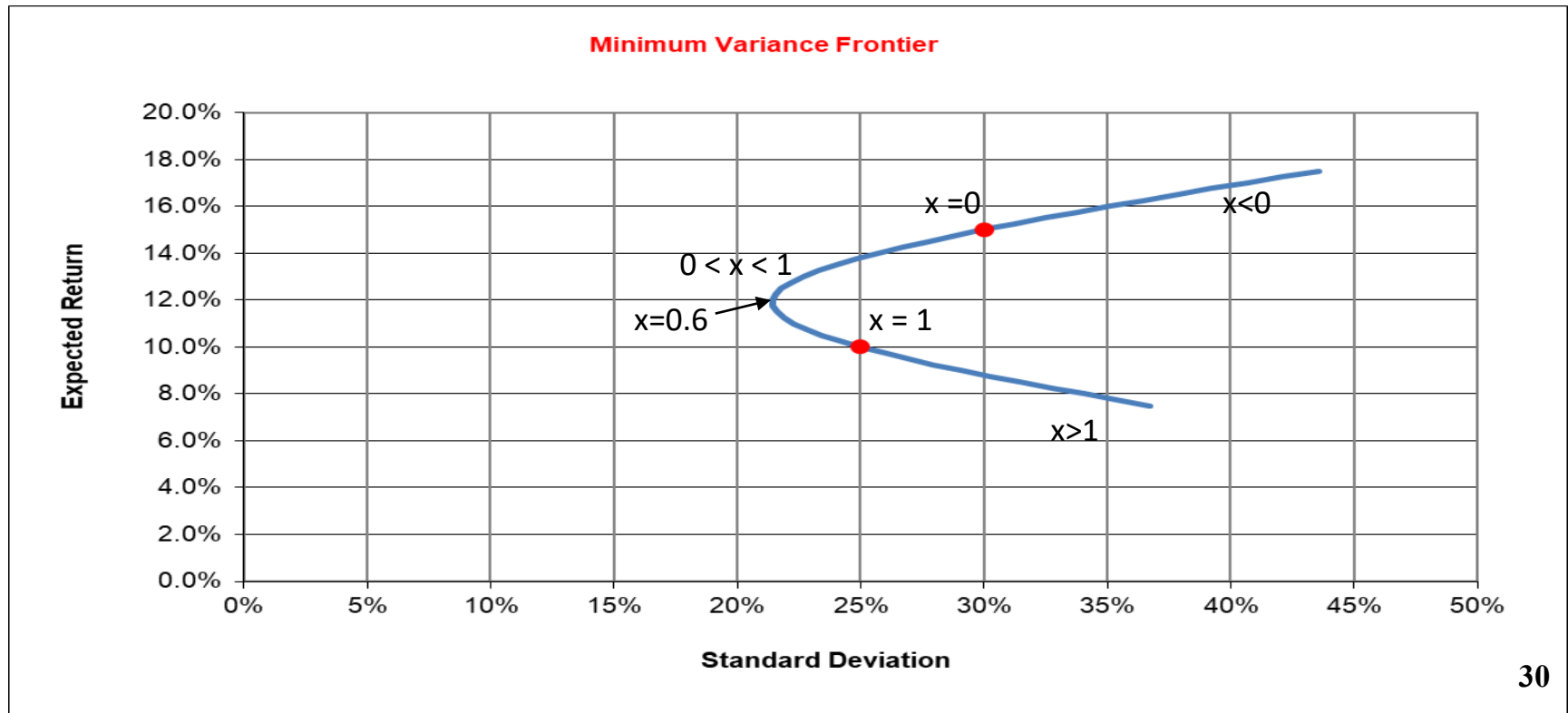
$$E[r_p] = x \times E[r_1] + (1-x) \times E[r_2] = x \times 0.10 + (1-x) \times 0.15 \quad 29$$

Intuition: Two Risky Assets Portfolio

- Its standard deviation is the square-root of its variance,

$$\text{Var}[r_p] = x^2 0.25^2 + (1 - x)^2 0.3^2 + 2x(1 - x)(\mathbf{0.25})(0.25)(0.3)$$

- Now vary x and see what happens to the portfolio expected return and standard deviation. We can plot these on a graph for various choices of x between -0.5 and 1.5 .



Intuition: Two Risky Assets Portfolio

- **Note:**

- $x = 1 \rightarrow (\sigma, \mu) = (0.25, 0.10)$; $x = 0 \rightarrow (\sigma, \mu) = (0.30, 0.15)$
- $x = 0.6 \rightarrow (\sigma, \mu) = (0.21, 0.12)$

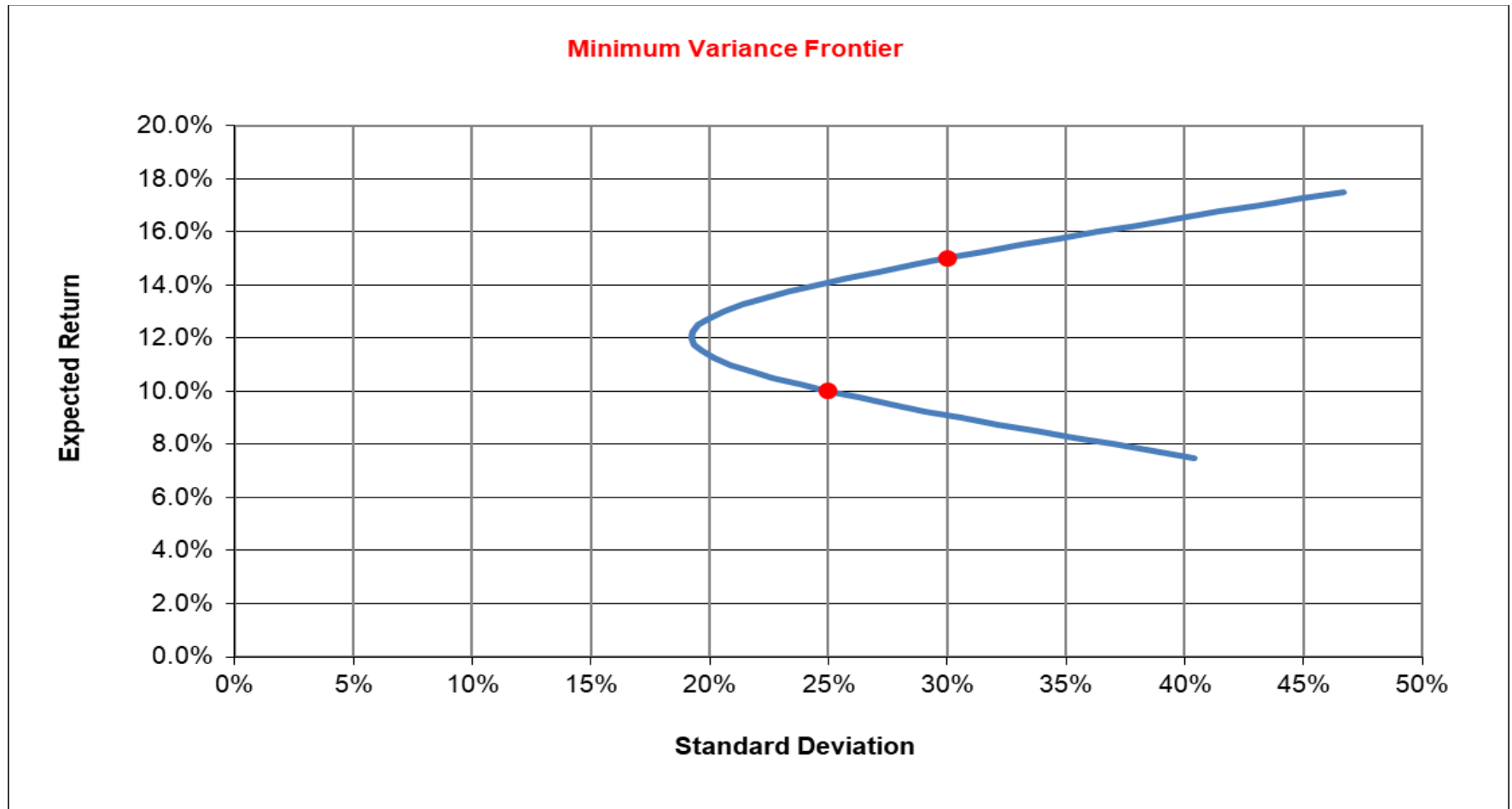
- **Points:**

- Between $0 < x < 1$ represents **long-positions** in both assets.
- Where $x > 1$ represents investments of more than 100% of the wealth in asset 1 and a **short position** in asset 2.
- Where $x < 0$ involves short positions in asset 1 with proceeds invested in asset 2.
- There is a wide range for x (i.e. $0.25 < x < 1$) in which the risk of portfolio (σ_p) is lower than either of the two assets: **Benefits of diversification!**
- **Bottom Line:** You will never choose a portfolio along the lower portion of the curve.

Intuition: Two Risky Assets Portfolio

Zero Correlation ($\rho_{12} = 0$)

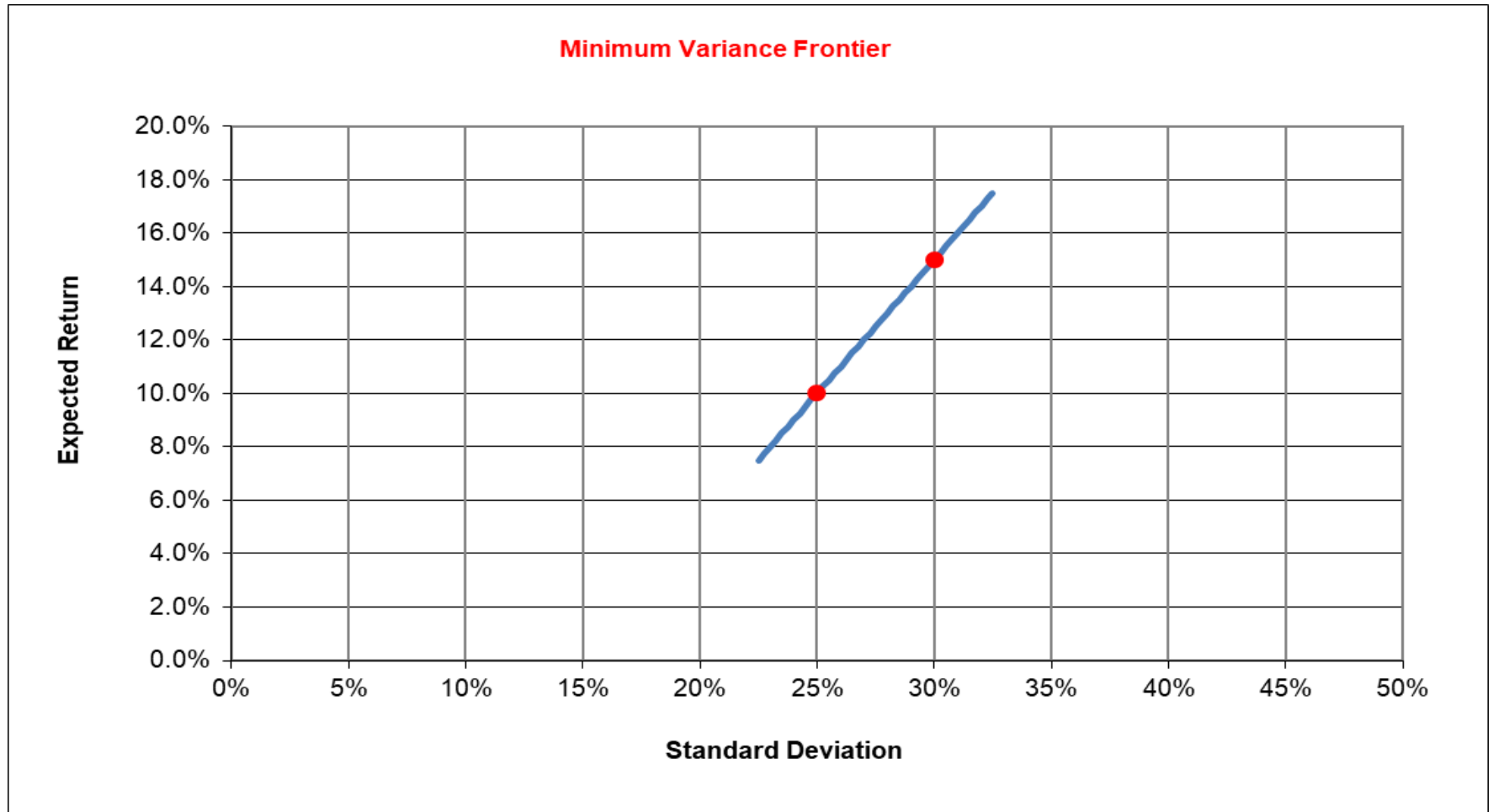
Showing The Effects of Diversification!



Intuition: Two Risky Assets Portfolio

Perfect Positive Correlation ($\rho_{12} = 1$)

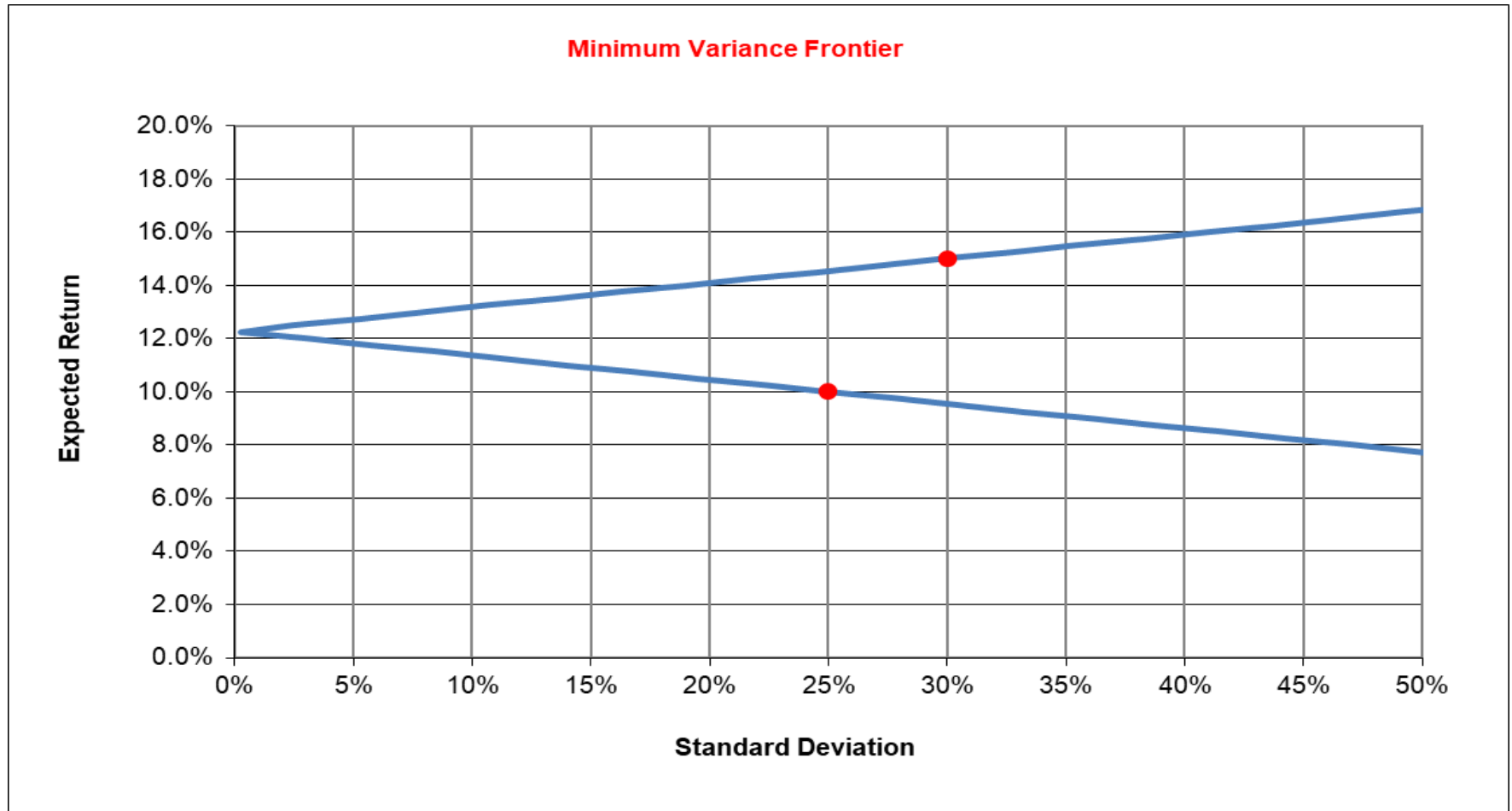
No Advantage Of Diversification!



Intuition: Two Risky Assets Portfolio

Perfect Negative Correlation ($\rho_{12} = -1$)

Advantages Of Diversification!!

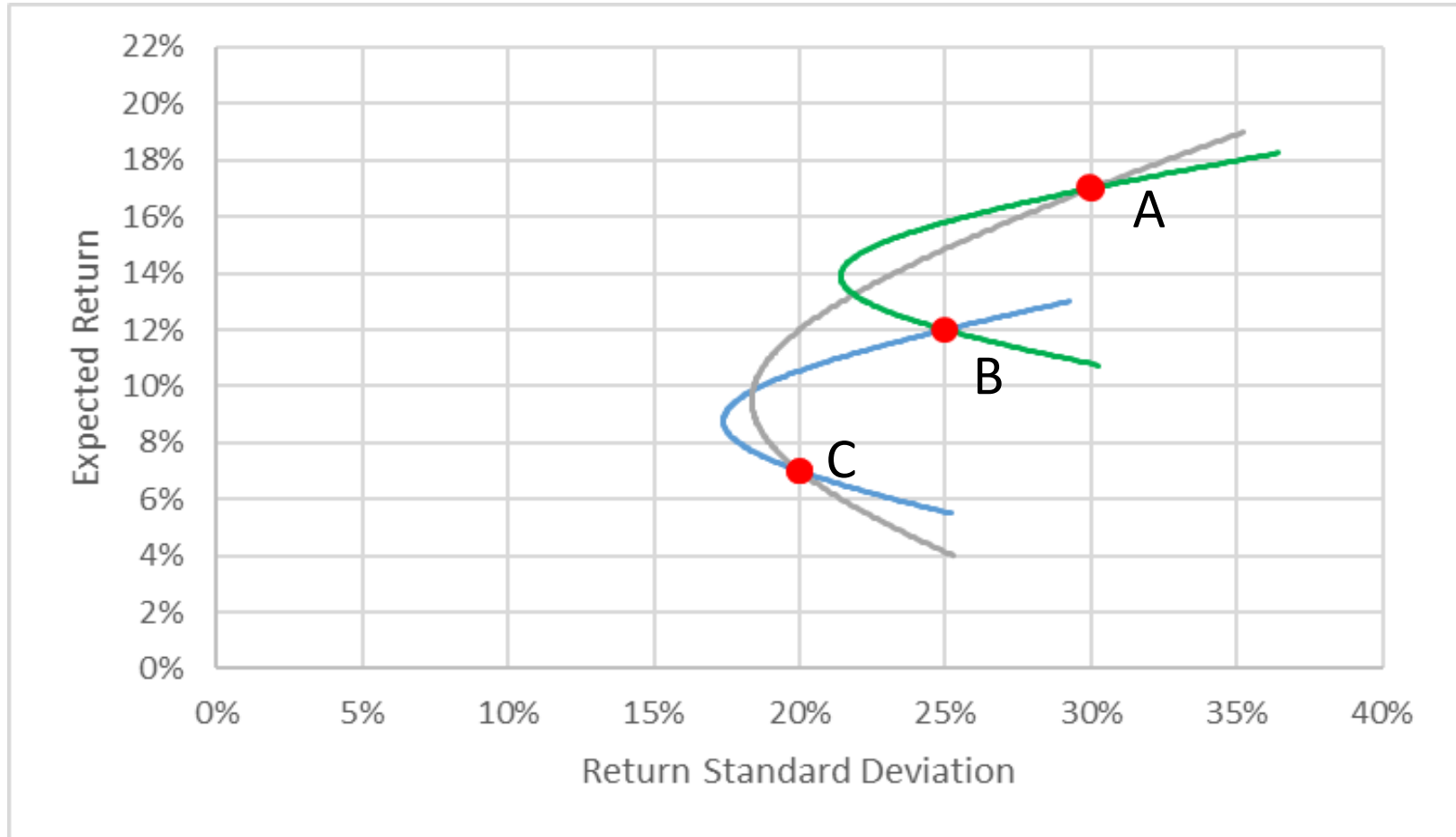


Generalizing the Intuition...

- Portfolio choice when an investor chooses a portfolio of a **single risky** and a **risk-free asset**.
 - CAL illustrates the available risk-return trade-off.
- Portfolio choice when an investor chooses a portfolio of **two risky assets**.
 - The Minimum Variance Frontier (MVF) illustrates the available risk-return trade-off.
 - Correlations determine the benefit of diversification.
- Now we generalize to cases where the investor can invest in:
 - Two or more risky assets.
 - A risk-free asset.
 - This is called **mean-variance portfolio analysis**

Generalizing the Intuition...

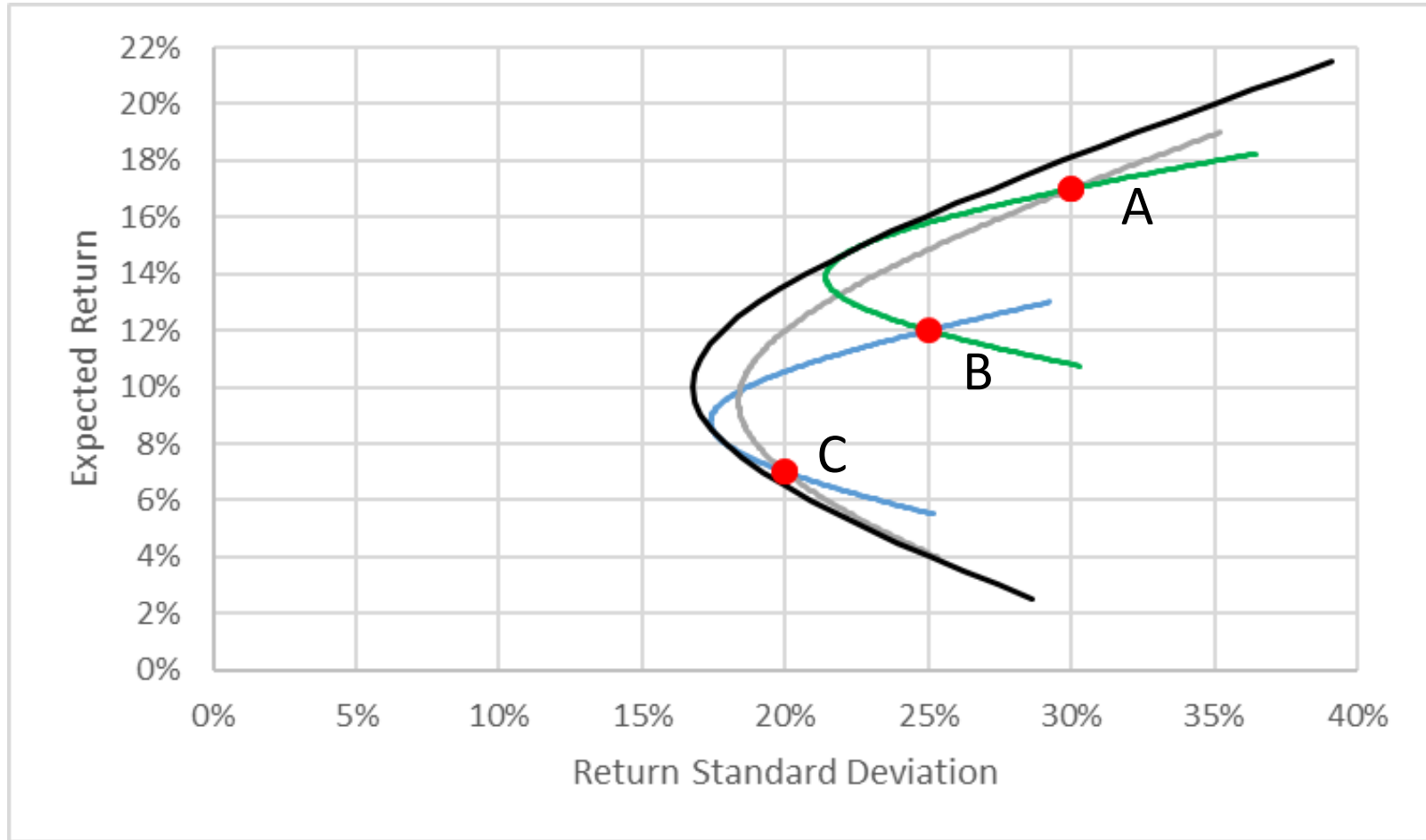
- Let's analyze the optimal portfolio problem when there are 3 risky assets: A, B and C.



- If we combine A and B, B and C, or A and C, we get three portfolio combinations AB, BC, AC.

Generalizing the Intuition...

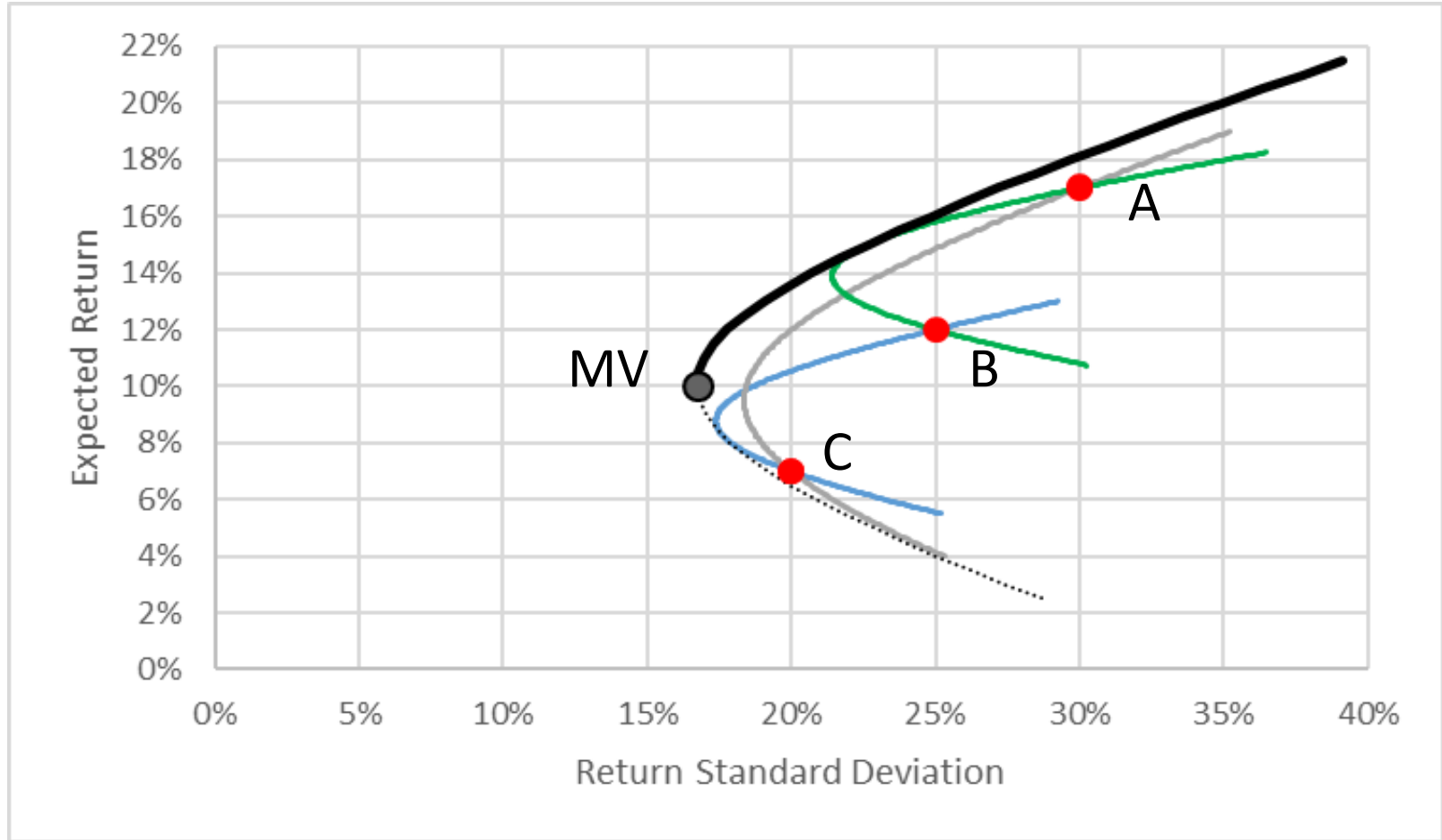
- This plot adds the **minimum-variance frontier (MVF)** of all the risky assets:



- The MVF of all three assets is the left boundary of the set of means and variances of the returns of all possible portfolios

Generalizing the Intuition...

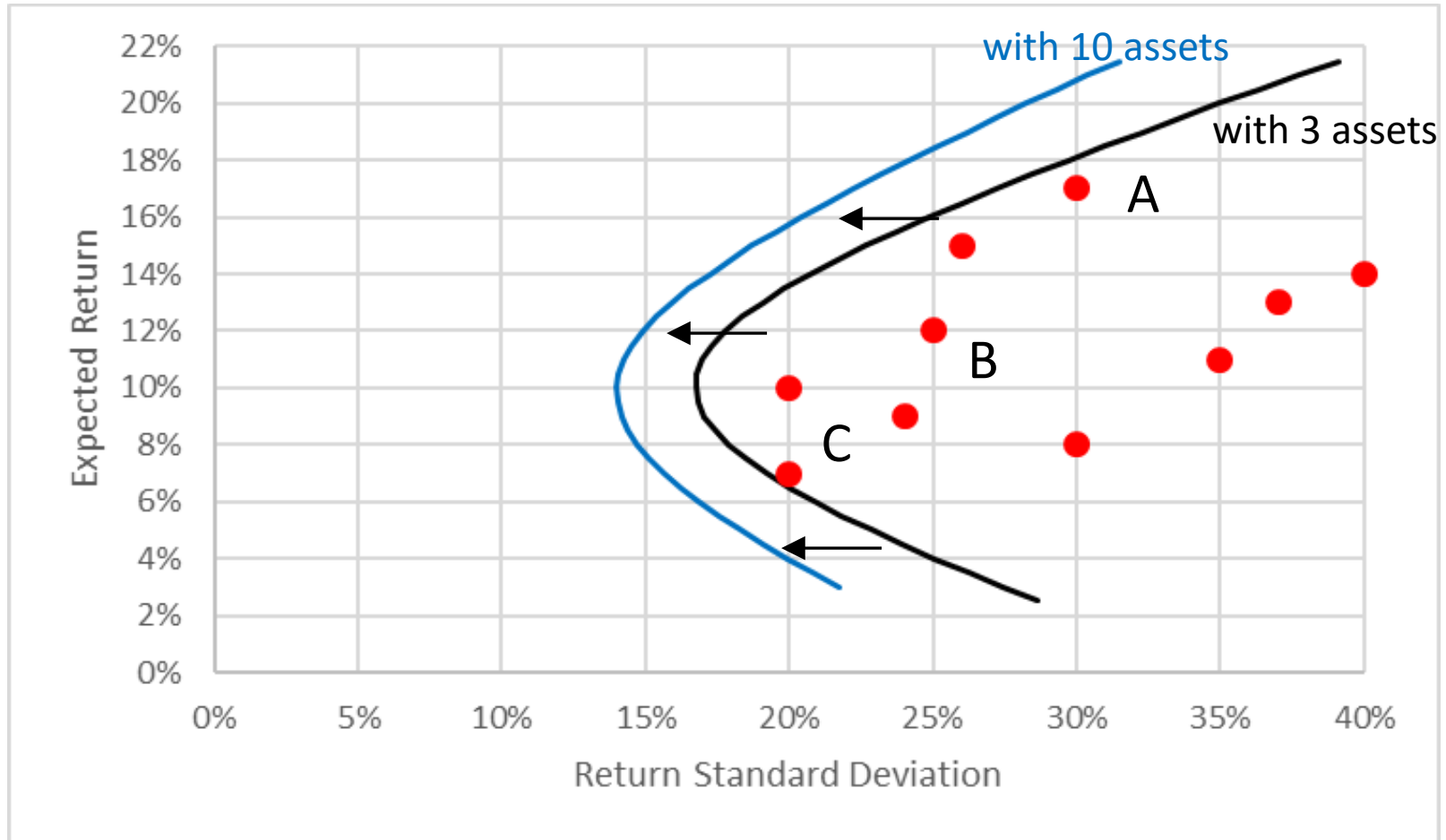
- This plot highlights the **Efficient Frontier**, which is the top part of the minimum-variance frontier of all the risky assets:



- The left most portfolio is the **Minimum Variance (MV) portfolio**:
The portfolio with the smallest risk.

Generalizing the Intuition...

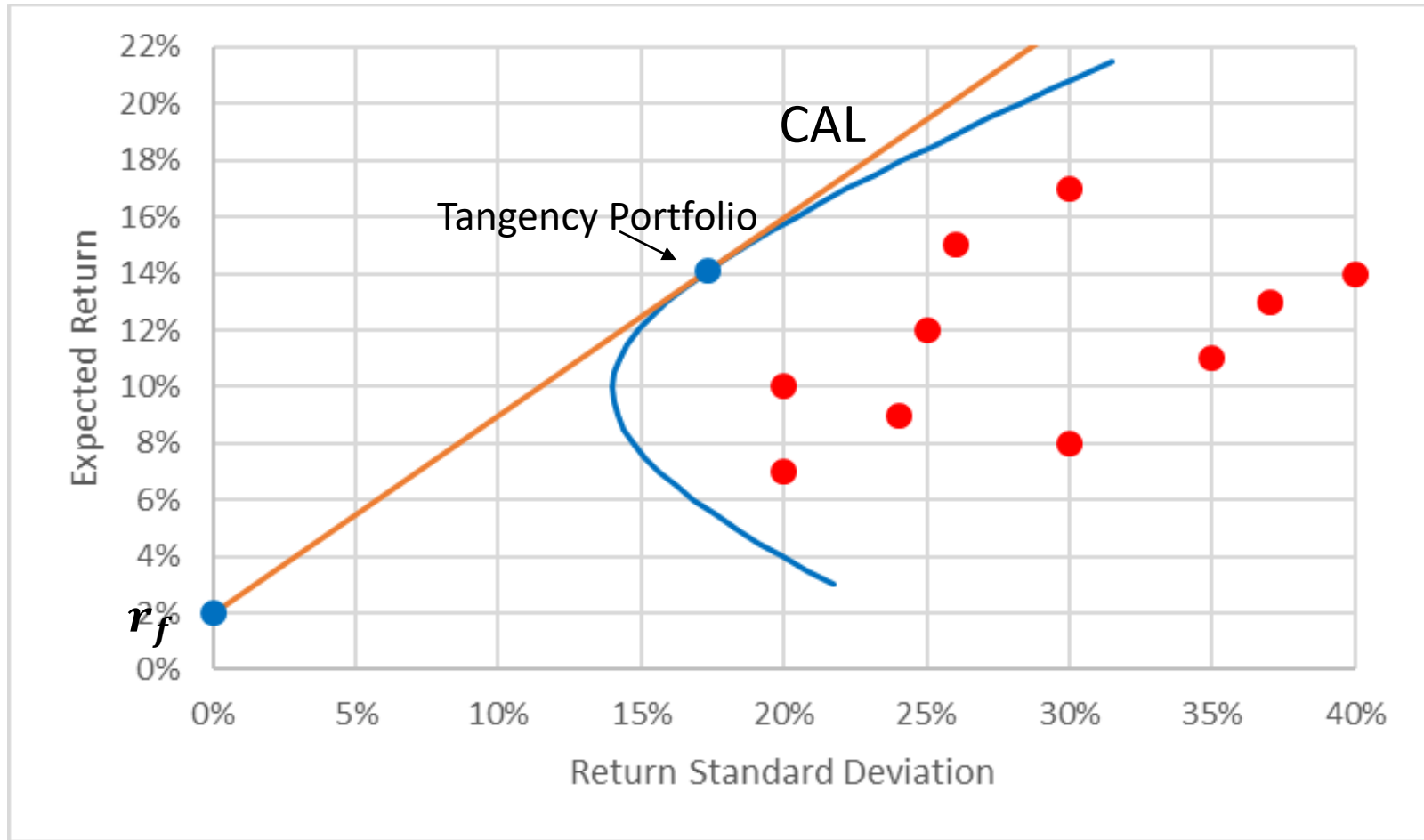
- With more assets, the **minimum-variance frontier (MVF)** shifts to the left as diversification opportunities increase:



- Note: the MVF shifts to the left, even if all new assets were *inside* the original MVF with only three assets

Suppose Add Risk-Free Asset Too...

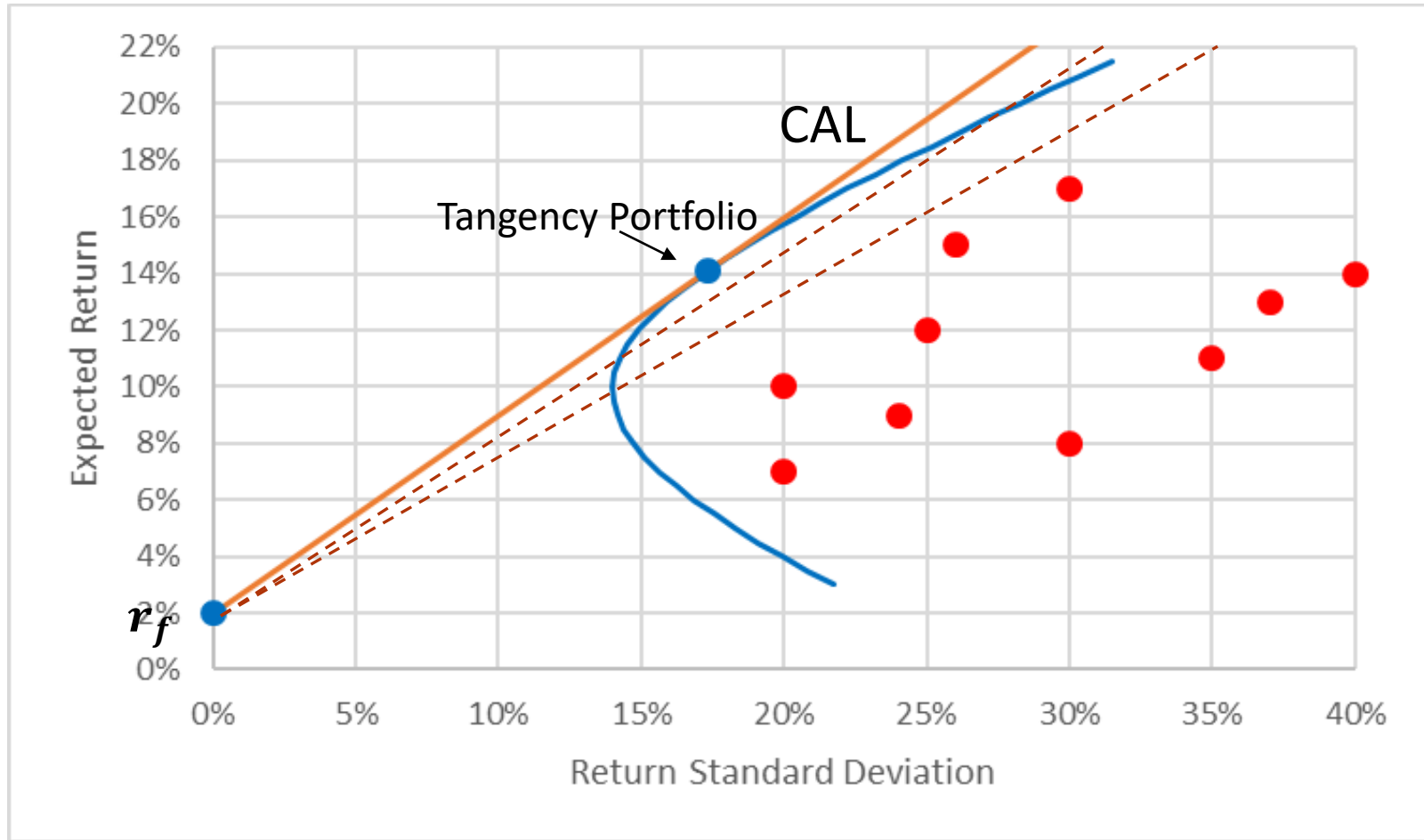
- Let's now plot the combination of three risky assets A, B, and C and a risk-free asset.



- The set of efficient combinations of the risk-free asset and the MVE portfolio is given by the straight line.

The Tangency Portfolio

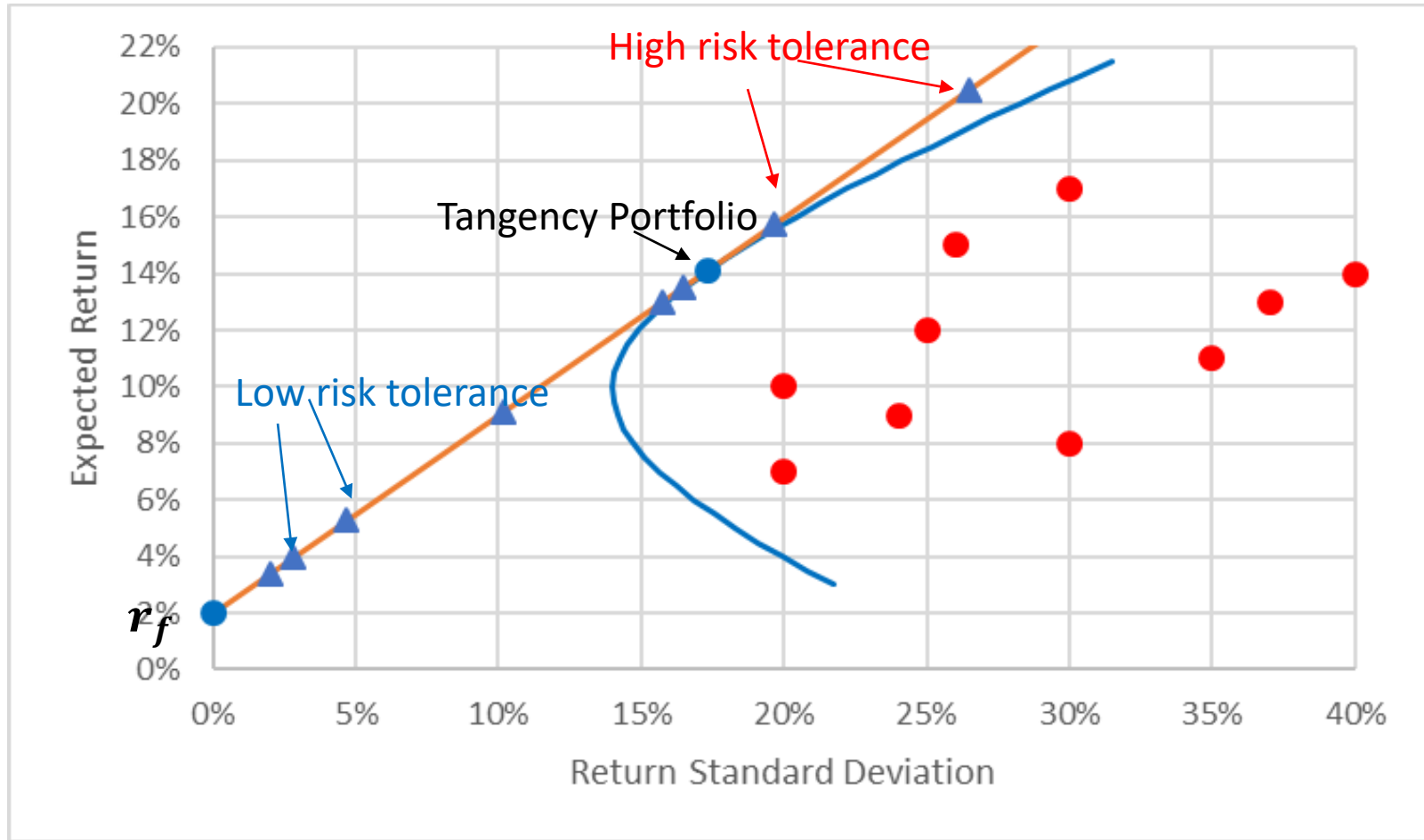
- The Tangency Portfolio is the portfolio of risky assets that maximizes the Sharpe ratio (i.e. slope of CAL)



- Any other combination of risky portfolio and the risk-free rate produces a lower Sharpe ratio (slope of line)

All Investors are on CAL

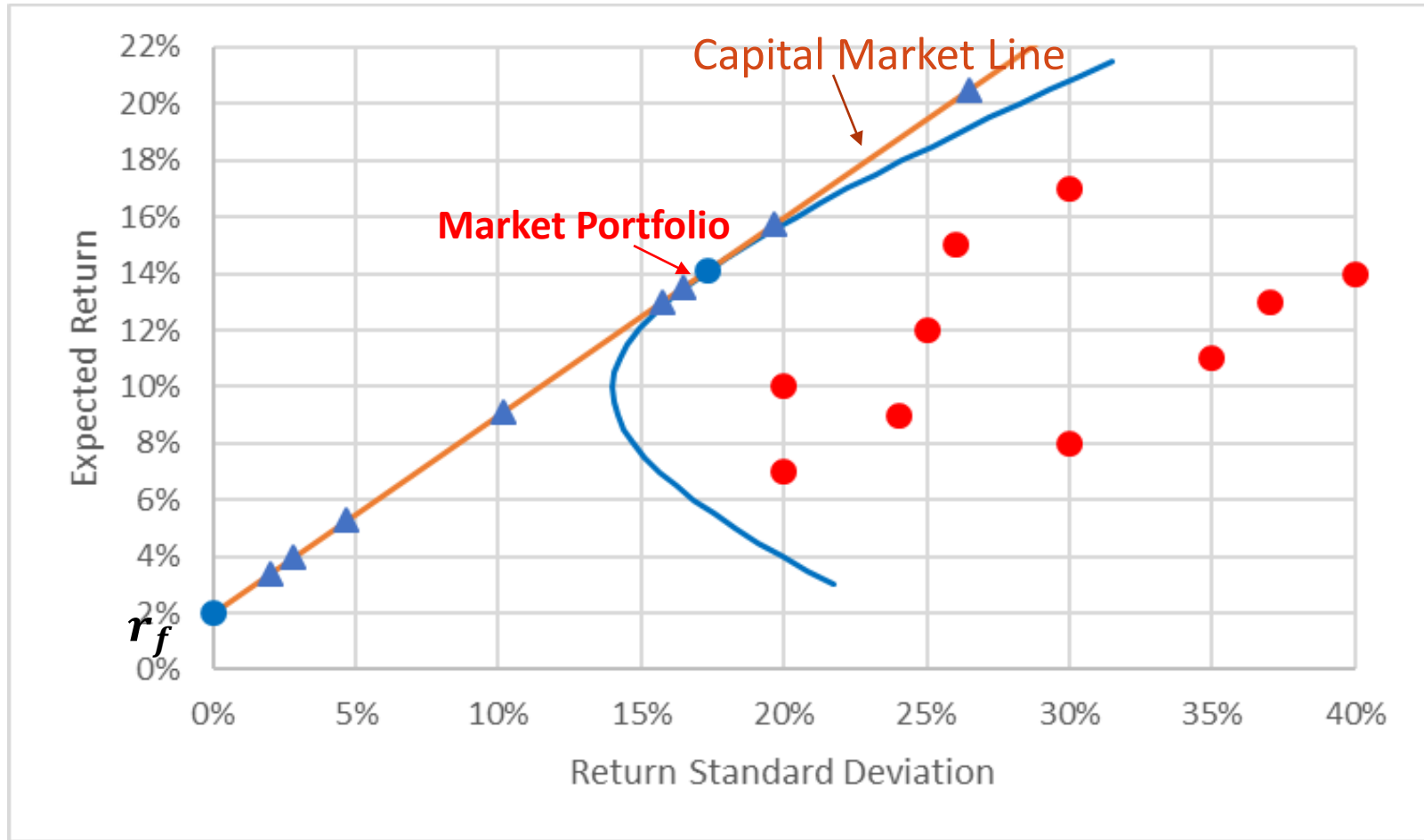
- All investors buy different combinations of risk-free assets and Tangency Portfolio (depending on their risk aversion)



- Low risk tolerance: $x < 1$ in stocks, $(1 - x)$ in T-bills
- High risk tolerance: $x > 1$ in stock, borrow $(1 - x < 0)$

Equilibrium

- Demand for risky securities (from investors) = Supply of risky assets
 - **Tangency Portfolio = Market Portfolio**



- The solid straight line is called **Capital Market Line**
 - Its slope is the Market Sharpe Ratio $[E(r_m) - r_f]/\sigma_m$

What do we do with all of this?

Portfolio Theory Implies...

- Need to think about “well-diversified” portfolios that lie on the mean variance efficient frontier
 - Want to know the contribution of an individual security to the risk of a well-diversified portfolio
- Many ways to think about measuring the “contribution” of an individual security to the risk of a well-diversified portfolio
 - The contribution can be captured by “market risk” of that security – i.e., how sensitive is that stock to market movements
 - Let us see a “preview” ...will discuss more in next lecture

Market Risk will be Measured By Beta

- The risk of a well-diversified portfolio depends on the market risk of the securities included in the portfolio.
- If you want to know the contribution of an individual security to the risk of a well-diversified portfolio:
 - You need to measure its **market risk** -- How sensitive is that stock to market movements.

$$\beta_{Stock\ i} = \frac{\sigma_{im}}{\sigma_m^2} = \frac{Cov(r_i, r_m)}{Var(r_m)}$$

- Stocks with:
 - $\beta > 1$ tend to **amplify** the overall movements of the market.
 - $0 < \beta < 1$ move in **same** direction as the market (but not as far).

Takeaways

- When investors hold portfolios of investments, their decisions about any individual asset depends on how much the asset **contributes** to:
 - Expected Return
 - Risk of the Portfolio.
- Measure of Risk:
 - Variation (variance) can be interpreted as a measure of risk.
- Risk-Return Trade-off:
 - Investors demand higher expected return for taking higher risk.
 - Graphically, we can see the trade-off from CAL
 - Sharpe Ratio, the slope of CAL, tells us how much expected return premium we get per 1% of standard deviation

Takeaways (cont.)

- Understanding the implications of **systematic** vs. **idiosyncratic risk** is one of the most important topics in finance.
 - Combining investments means the covariance between the investments matters for risk and return of the portfolio.
 - As a result, the overall portfolio standard deviation is lower than any of the individual assets' standard deviations.
- **Mean-Variance Portfolio** Analysis:
 - Investors care about expected return and variance of portfolio
 - Market risk accounts for most of the risk of a well-diversified portfolio.
- **Portfolio Theory** tells us that investors hold the same portfolio of risky assets no matter what their tolerance for risk
 - Contribution of an individual security to overall risk is captured by market risk – i.e., the **Beta**

Appendix

Appendix

Obtaining the Efficient Frontier in Practice

Obtaining the Efficient Frontier - 1

- How do we obtain the efficient frontier, in practice?

- Inputs:

- Assets expected excess returns: $\mu_1, \mu_2, \dots, \mu_n$
- Assets variances: $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$
- Assets' pairwise correlations: $\rho_{1,2}, \rho_{1,3}, \dots, \rho_{i,j} \dots \rho_{n-1,n}$
 - Where $\rho_{i,j} = \text{Corr}(r_i, r_j)$

- Consider a set of portfolio weights: $x_1, x_2, \dots, x_n$

- Recall $x_1 + x_2 + \dots + x_n = 1$

- Portfolio expected return:

$$E[r^p] = x_1\mu_1 + x_2\mu_2 + \dots + x_n\mu_n$$

- Portfolio Variance:

$$V[r^p] = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + \dots + x_n^2\sigma_n^2 + \\ + x_1x_2\sigma_1\sigma_2\rho_{1,2} + x_1x_3\sigma_1\sigma_3\rho_{1,3} + \dots + x_{n-1}x_n\sigma_{n-1}\sigma_n\rho_{n-1,n}$$

Obtaining the Efficient Frontier- 2

- Find the weights to minimize the variance of the portfolio, for a given level of portfolio expected return.
 - That is, define a target portfolio excess return μ^{Target}
 - Find portfolio weights to solve

$$\min_{x_1, x_2, \dots, x_n} V[r^p] \quad \text{subject to}$$

$$E[r^p] = \mu^{Target} \text{ and } x_1 + x_2 + \dots + x_n = 1$$

- The solution gives a set of portfolio weights $x_1, x_2, \dots, x_n$
- Repeat the above for many μ^{Target} .
 - ➔ trace out the whole frontier, i.e. pairs of μ^{Target} and the corresponding minimized standard deviation $\sigma_{min}(\mu^{Target})$
- Luckily, it turns out there is a solution to the minimization problem

Obtaining the Efficient Frontier - 3

- Define:

- C = Variance Covariance Matrix =
$$\begin{bmatrix} \sigma_1^2, \sigma_{1,2}, \sigma_{1,3} \dots \sigma_{1,n} \\ \sigma_{2,1}, \sigma_2^2, \sigma_{2,3} \dots \sigma_{2,n} \\ \vdots \quad \vdots \quad \vdots \quad \ddots \quad \vdots \\ \sigma_{n,1}, \sigma_{n,2}, \sigma_{n,3} \dots \sigma_n^2 \end{bmatrix}$$

- $\vec{\mu} = (\mu_1, \mu_2, \dots, \mu_n)'$

- $a = \vec{\mu}' C^{-1} \vec{\mu}; b = \vec{\mu}' C^{-1} \mathbf{1}; c = \mathbf{1}' C^{-1} \mathbf{1}; d = a - \frac{b^2}{c};$

- Then, denoting the expected excess return μ^{Target} simply as μ , the minimum variance is:

$$\sigma_{min}^2(\mu) = \frac{a}{dc} - 2 \frac{b}{dc} \mu + \frac{1}{d} \mu^2$$

Obtaining the Efficient Frontier - 4

- The corresponding portfolio weights are (as vector)

$$\vec{x} = \vec{q}^{MV} \frac{(a - b\mu)}{d} + \vec{q}^{TG} \frac{(\mu b - b^2/c)}{d}$$

where

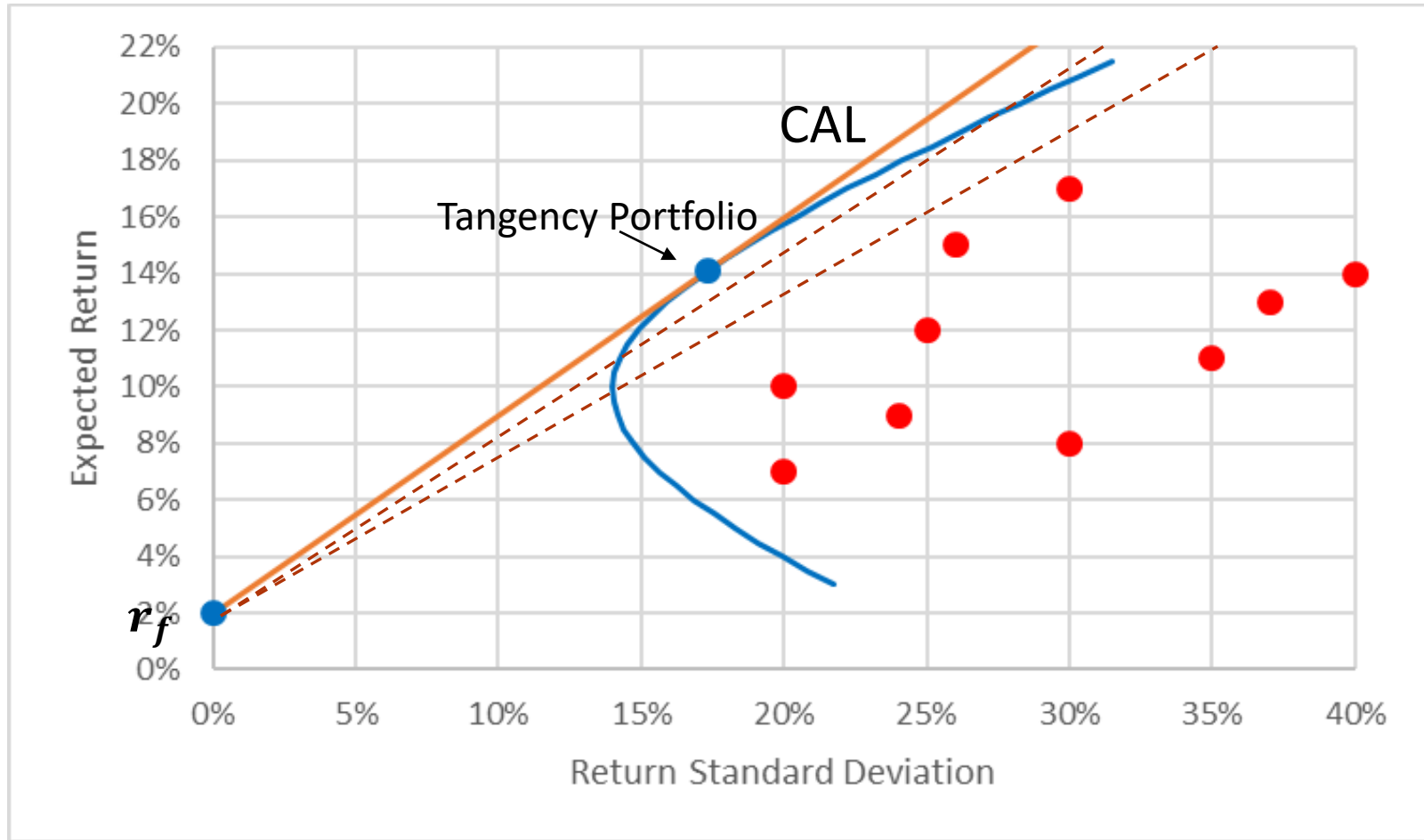
$\vec{q}^{MV} = C^{-1} \mathbf{1}/c = \text{Minimum Variance Portfolio}$

$\vec{q}^{TG} = C^{-1} \vec{\mu}/b = \text{Tangency Portfolio}$

- Recall, the tangency portfolio $\vec{q}^{TG}$ is the one that maximizes the Sharpe ratio

The Tangency Portfolio

- The Tangency Portfolio is the portfolio of risky assets that maximizes the Sharpe ratio (i.e. slope of CAL)

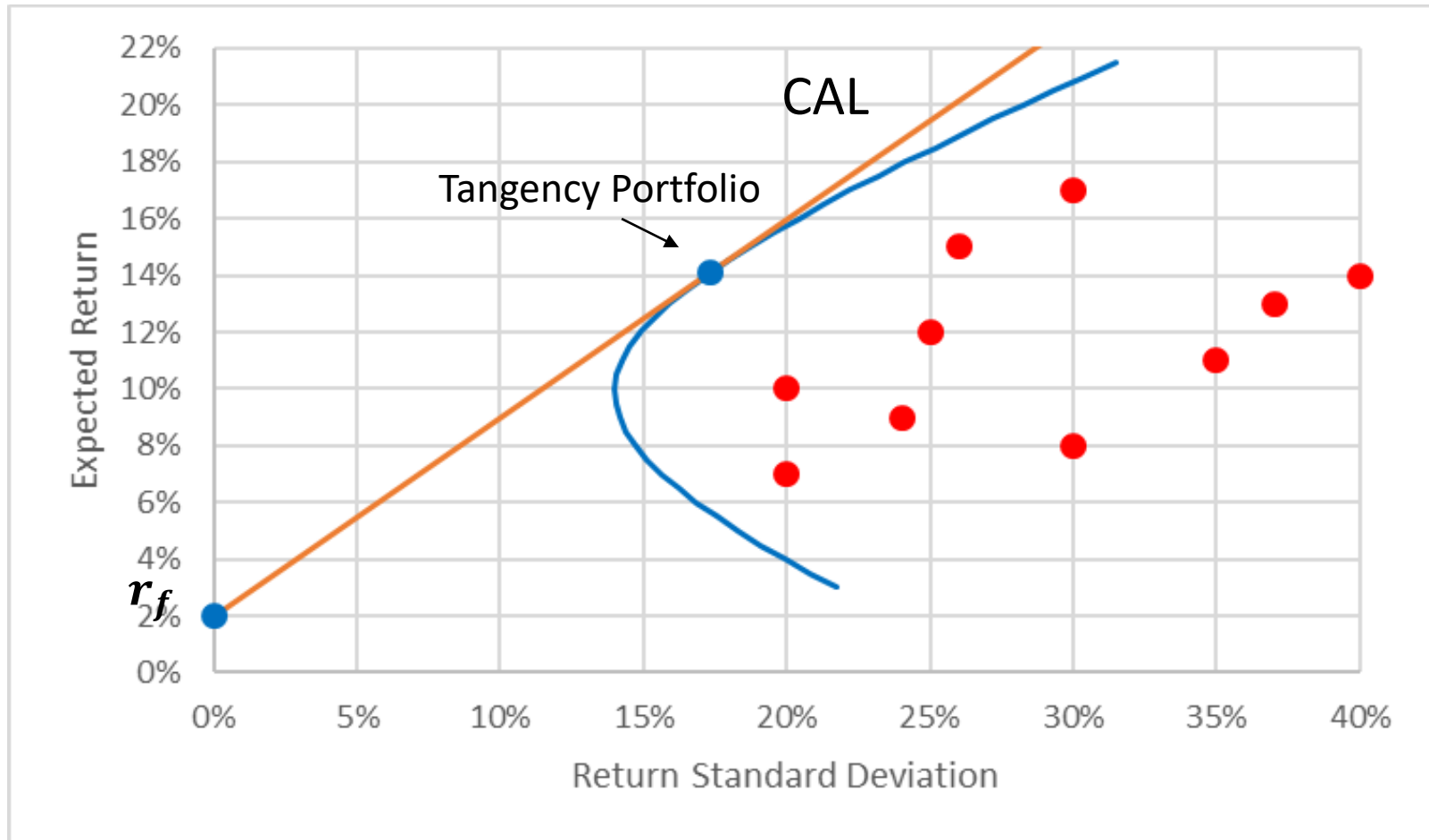


- Any other combination of risky portfolio and the risk-free rate produces a lower Sharpe ratio (slope of line)

Maximization of Utility

The Tangency Portfolio

- How do we pick the optimal point on the tangency portfolio?
 - Let $E[r_T]$ and σ_T be the mean and standard deviation on tangency portfolio



Review of Probability and Statistics

Random Variables: Probability Distributions

- Random variables are the fundamental elements of probability theory and are used to characterize **uncertainty**.
 - Random variables take on different values in different states of nature.
- A random variable can be described entirely by its probability distribution.
- **Example:** The face that shows when you toss a coin is a random variable.
 - The possible realizations are “head” and “tail”.
 - If “heads” is twice as likely as “tails”, then the probability distribution is:

Realization	Probability
Head	$2/3$
Tail	$1/3$

Random Variables: Probability Distributions

- We usually define the realizations of a random variable as numerical values.
 - If Gambler 1 bets \$1 on heads and Gambler 2 bets \$1 on tails.
 - The random variable that characterizes the winnings and losses of Gambler 1 has the probability distribution:

W_1 = Gambler 1's Winnings

Realization	Probability
Head	$2/3$
Tail	$1/3$

- **Note:** This is a simple random variable because it just has two possible realizations.
 - Uncertainty is more complicated in real life.

Returns *Are* Random Variables

- The return to a share of common stock can also be described as a random variable.

$$P_0 = \frac{Div_1 + P_1}{1 + r}$$

- The security's return over the period is given by:

$$r = \frac{Div_1}{P_0} + \left(\frac{P_1 - P_0}{P_0} \right)$$

- r is a **random variable** because Div_1 and P_1 are **not** known.
 - r can take on any value from -1 to infinity
- In principle, r can be described by its probability distribution.
 - Describing the entire probability distribution is not feasible so we focus on a few summary statistics

Expectation and Standard Deviation

- A general probability distribution for the return to a stock, R , is given by:

Realization	Probability
r_1	p_1
r_2	p_2
.	.
.	.
r_n	p_n

- Only restriction is that probabilities must **add** to 1.
- The **Expectation** of R is:

$$\mu_R = E[R] = p_1 \times r_1 + p_2 \times r_2 + \dots + p_n \times r_n$$

- We need to weight outcomes by their probabilities since more likely realizations would occur more often in many trials.

Expectation and Standard Deviation (cont.)

- To define how far R tends to depart from its expectation, we need a measure of distance.
 - We use **variance** as a measure of distance.
 - **Question:** Why not $(r_i - \mu_R) = (r_i - E_R)$?

- We define the variance of R by:

$$\sigma_R^2 = \text{Var}[R] = p_1 (r_1 - \mu_R)^2 + p_2 (r_2 - \mu_R)^2 + \dots + p_n (r_n - \mu_R)^2$$

- This is the average deviation of R from its expectation in squared units.
- We take the square-root to get the standard deviation:

$$\sigma_R = \text{Std. Dev}[R] = \sqrt{\text{Var}(R)} = \sqrt{\sigma_R^2}$$

Covariance and Correlation

- The **covariance** describes the degree to which two random variables move in the same direction.
 - This is important for finance because the risk of a portfolio depends on the covariance between the securities in the portfolio.
- **Example:** Rolls Royce vs. Generic Food Items
 - Demand for Rolls Royce cars (and the profitability of firms that sell them) increases in “booms” and decreases in “busts”.
 - Demand is Pro-cyclical.
 - Demand for Generic Food Items (and the profitability of firms that sell them) decreases and is unprofitable in “booms”, and increases in “busts”.
 - Demand is Counter-cyclical.

Covariance and Correlation (cont.)

- The returns to Rolls Royce and Generic Food Items are **negatively** related.
 - They both move in response to the state of the economy in **opposite** directions.
- The return to Rolls Royce and Generic Food can be described by the *joint probability distribution*.

Generic \ Rolls Royce	Boom +50%	Bust -20%
Boom -10%	$\frac{3}{4}$	0
Bust +25%	0	$\frac{1}{4}$

- **Note:** The probabilities are for **joint** events (*i.e.* events that specify the realization of both random variables) and these probabilities add to one.

Covariance and Correlation (cont.)

- In general, the joint probability distribution between two random variables (*i.e.* the random returns of two different stocks, R and S) is given by:

R \ S	s₁	s₂	...	s_n
r₁	p ₁₁	p ₁₂	...	p _{1n}
r₂	p ₂₁	p ₂₂	...	p _{2n}
...	...	...	...	...
r_n	p _{n1}	p _{n1}	...	p _{nn}

- The covariance is a summary statistic. It is defined as $\sigma_{R,S}$:

$$\text{Cov}[R, S] = p_{11} (r_1 - \mu_R) (s_1 - \mu_S) + p_{12} (r_1 - \mu_R) (s_2 - \mu_S) + \dots + p_{nm} (r_n - \mu_R) (s_m - \mu_S)$$

- **Note:** This is the probability-weighted average of all possible joint deviations of the random variables from their respective means.

Covariance and Correlation (cont.)

- If R and S tend to move **together**, $\text{Cov}[R,S]$ will be **positive**.
- If R and S tend to move **opposite**, $\text{Cov}[R,S]$ will be **negative**.
- **Correlation** is a normalized measure of covariance.
 - It is sometimes easier to think about because it is unitless.
 - It takes values in the interval $[-1,1]$.
 - Random variables that have a correlation of:
 - 1 are perfectly correlated.
 - -1 are perfectly negatively correlated.
 - 0 are uncorrelated.

$$\text{Corr}[R, S] = \rho_{R,S} = \frac{\text{Cov}[R, S]}{\sigma_R \times \sigma_S}$$

Summary Statistics for Generic and Rolls Royce

- **Expected Returns:**

- $E[r_{\text{Rolls}}] = 0.75(0.5) + 0.25(-0.2) = 0.325$
- $E[r_{\text{Generic}}] = 0.75(-0.1) + 0.25(0.25) = -0.0125$

- **Variance:**

- $\text{Var}[r_{\text{Rolls}}] = 0.75(0.5 - 0.325)^2 + 0.25(-0.2 - 0.325)^2 = 0.0919$
 - $\sigma[r_{\text{Rolls}}] = 0.303$
- $\text{Var}[r_{\text{Generic}}] = 0.75(-0.1 - (-0.0125))^2 + 0.25(0.25 - (-0.0125))^2 = 0.0230$
 - $\sigma[r_{\text{Generic}}] = 0.152$

- **Covariance/Correlation:**

- $\text{Cov}[r_{\text{Rolls}}, r_{\text{Generic}}] = 0.75(0.5 - 0.325)(-0.1 - (-0.0125))$
 $+ 0.25(-0.2 - 0.325)(0.25 - (-0.0125)) = -0.0459$
- $\text{Corr}[r_{\text{Rolls}}, r_{\text{Generic}}] = -0.0459 / [(0.303)(0.152)] = -1$

- **Bottom Line:** Returns are perfectly **negatively** correlated!

Rules On Expectations, Variance, And Covariance

Let c be a constant (ie: c takes a constant value and does not vary)

- **Properties of Expected Value (Mean)**

- $E(x + c) = E(x) + c$
- $E(a \times x) = a \times E(x)$
- $E(x + y) = E(x) + E(y)$

- **Properties of Variance**

- $\text{Var}(a \times x) = a^2 \times \text{Var}(x)$
- $\text{Var}(x + y) = \text{Var}(x) + \text{Var}(y) + 2 \text{Cov}(x, y)$
- $\text{Var}(a \times x + b \times y) = a^2 \times \text{Var}(x) + b^2 \times \text{Var}(y) + 2ab \times \text{Cov}(x, y)$
- $\text{Var}(x + c) = \text{Var}(x)$

- **Properties of Covariance**

- $\text{Cov}(x, y) = \text{Cov}(y, x)$
- $\text{Cov}(a \times x, y) = a \times \text{Cov}(x, y)$
- $\text{Cov}(x, c) = 0$

A “Short” Note on Short-Selling

- How do we get negative portfolio weights? If you have a negative position in an asset, you “owe” it, or you have a **short position**. To short sell an asset:
 1. Borrow the asset from someone who owns it, then sell it into the market at its current price, say P_0 .
 2. Later close out the short position by buying the security for P_1 and returning it to the lender.
- The cash flows from these transactions are:
 - Time 0: $+ P_0$ (from selling the asset)
 - Time 1: $- P_1$ (from buying back the asset)
- The return to short-selling is $(-P_1 + P_0)/P_0 = -(P_1 - P_0)/P_0$
 - The short-seller *pays* the asset’s random return.

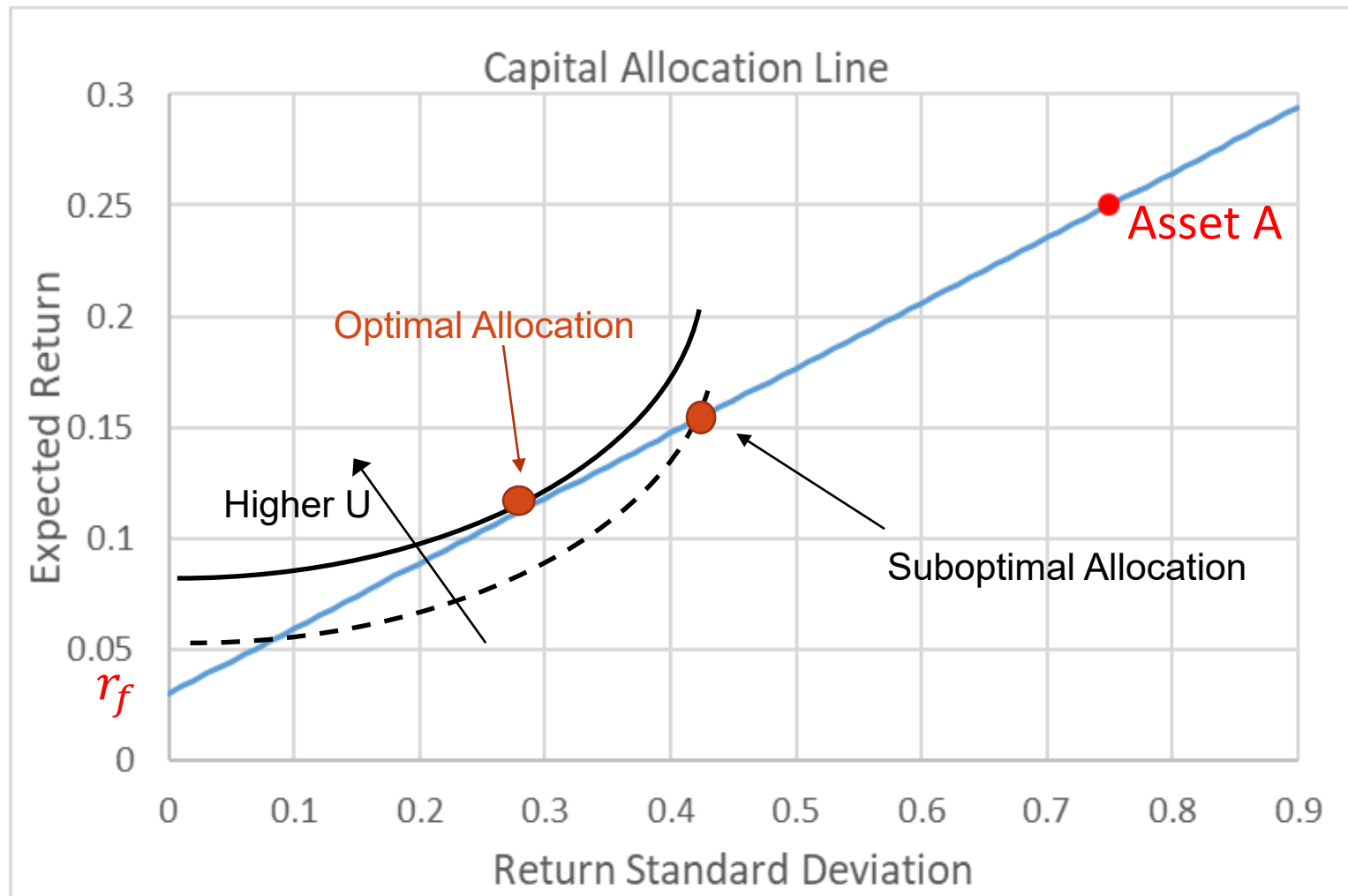
Risk Aversion and Portfolio Holdings

Optimal Mix between Stocks and Bonds

- **Question:** How should investors choose their risk-return combination?
- Assume investors:
 - Prefer high returns, everything else equal
 - Prefer low risk, everything else equal
- Simple model: **Mean-Variance preferences**
- Utility (happiness): $U(E, \sigma) = E - \frac{1}{2}A\sigma^2$
 - $E = E[r_p]$ = Expected Portfolio Return
 - $\sigma^2 = V[r_p]$ = Variance of Portfolio Return
 - A = degree of aversion to risk
- **Objective:** Choose weight in risky asset x to maximize $U(\mu, \sigma)$ given the risk-return combinations on CAL.

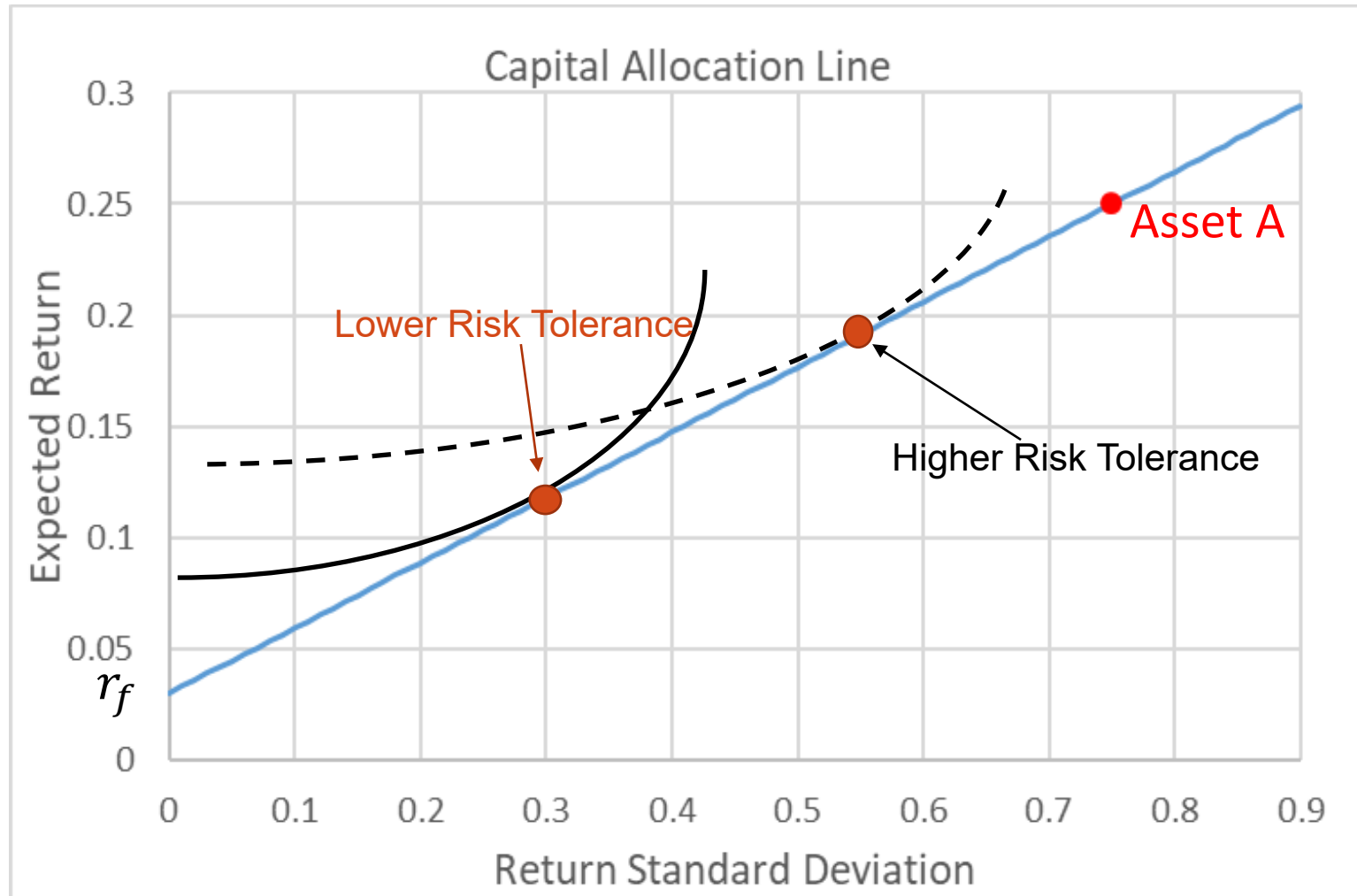
Optimal Mix between Stocks and Bonds

- Indifference Curve (IC): pairs (E, σ) with $U(E, \sigma) = u_0$
- Optimal allocation = tangency point of IC to CAL



Optimal Mix between Stocks and Bonds

- Higher risk tolerance → higher weight to risky asset A



Optimal Mix between Stocks and Bonds

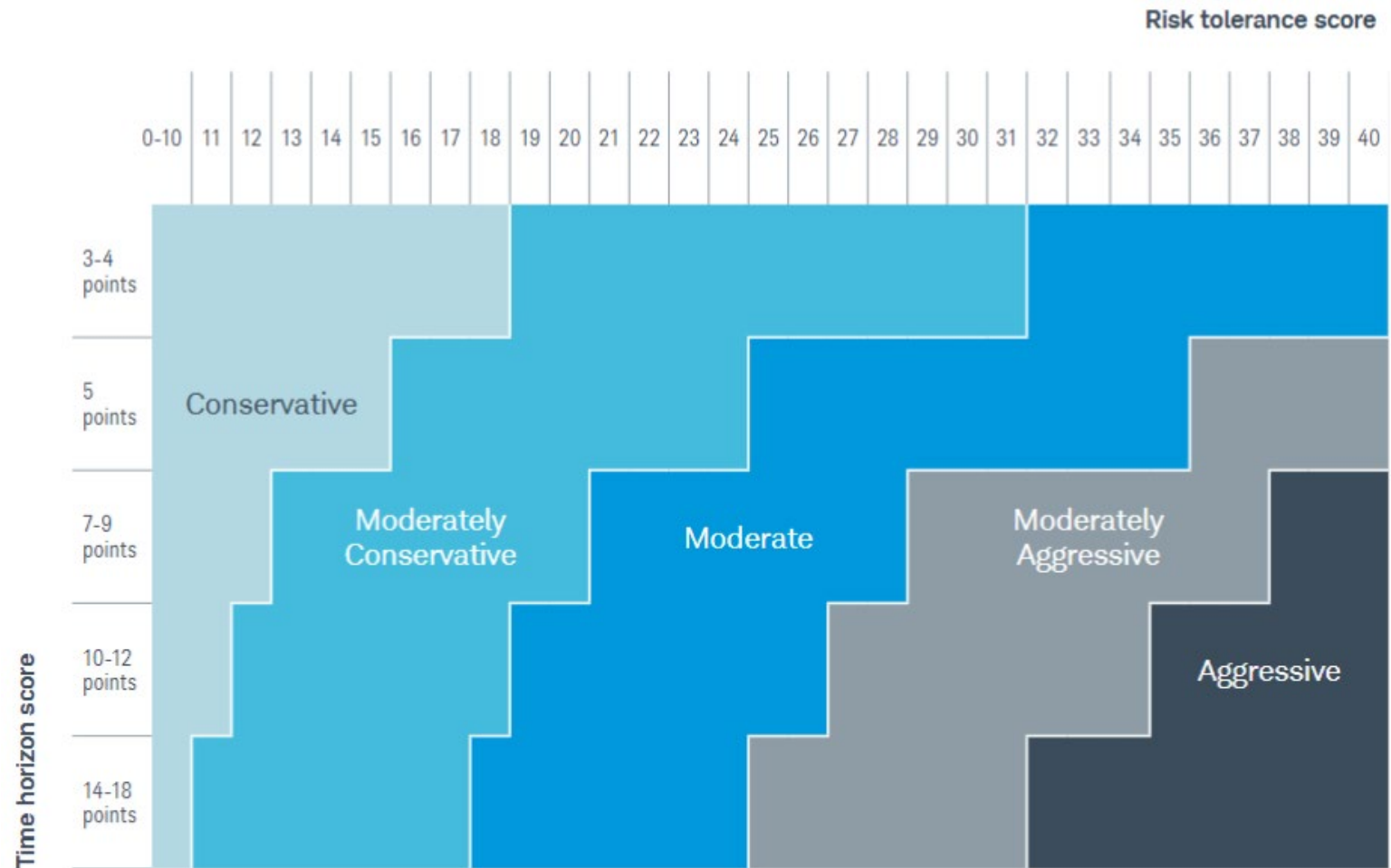
- **Result:** Below I show that for mean-variance preferences the optimal allocation to the risky asset is:

$$x^* = \frac{1}{A} \frac{E[r_A] - r_f}{\sigma_A^2}$$

- Intuitively, higher risk premium of the risky asset ($E[r_A] - r_f$) increases allocation to the risky asset, while higher risk (σ_A^2) or aversion to risk (A) decreases it.
- **Example:** Let $E[r_A] - r_f = 6\%$, $\sigma_A = 15\%$, then
 - If $A = 2$, then $x^* = 133.3\%$
 - If $A = 4$, then $x^* = 66.6\%$
 - If $A = 8$, then $x^* = 33.3\%$
- How do we determine the risk aversion A ?

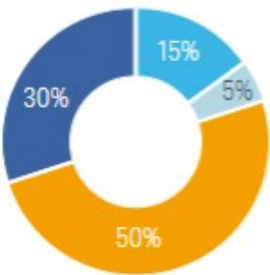
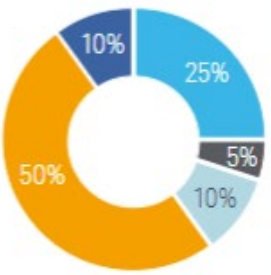
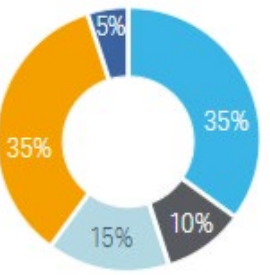
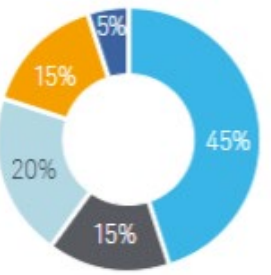
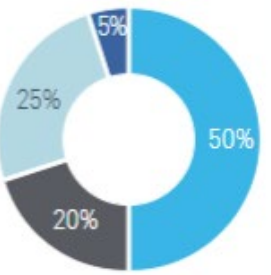
Optimal Mix between Stocks and Bonds

- In practice, questionnaires help elicit risk tolerance
- Including time horizon limits risk of low savings at retirement



Optimal Mix between Stocks and Bonds

- Translate risk tolerance/time horizon in investment strategy
- Equity vs cash/fixed income and types of equity [later]

Conservative allocation	Moderately conservative	Moderate allocation	Moderately aggressive	Aggressive allocation
Average annual return: 7.0% Best year: 22.8% Worst year: -9.5%	Average annual return: 8.3% Best year: 27.0% Worst year: -13.3%	Average annual return: 9.0% Best year: 30.9% Worst year: -20.9%	Average annual return: 9.6% Best year: 34.4% Worst year: -29.5%	Average annual return: 9.9% Best year: 39.9% Worst year: -36.0%
For investors who seek current income and stability and are less concerned about growth.	For investors who seek current income and stability, with modest potential for increase in the value of their investments.	For long-term investors who don't need current income and want some growth potential. Likely to entail some fluctuations in value, but presents less volatility than the overall equity market.	For long-term investors who want good growth potential and don't need current income. Entails a fair amount of volatility, but not as much as a portfolio invested exclusively in equities.	For long-term investors who want high growth potential and don't need current income. May entail substantial year-to-year volatility in value in exchange for potentially high long-term returns.
				

■ Large-Cap Equity
 ■ Small-Cap Equity
 ■ International Equity
 ■ Fixed Income
 ■ Cash Investments

(Source: Charles Schwab <https://www.schwab.com/resource/investment-questionnaire>)

Obtaining the Optimal Allocation - 1

- The investor has mean-variance utility function

$$U(E, \sigma) = E - \frac{1}{2}A\sigma^2$$

- where E is the expected return on the investment, σ is its risk (standard deviation) and A is the degree of risk aversion.
- As in microeconomics, this is a standard maximization of utility over two “goods” (actually, one “good” E and one “bad” σ) on a budget constraint. The budget constraint is the Capital Allocation Line CAL between the risk free rate r_f and the Tangency portfolio. Recall the CAL (budget constraint) is

$$E = r_f + \left(\frac{E[r_T] - r_f}{\sigma_T} \right) \sigma$$

- Substitute into the utility function and take the first order conditions with respect to σ to find the “target” risk σ^*

Obtaining the Optimal Allocation - 2

- We have

$$\begin{aligned} \max_{\sigma} U(E, \sigma) &= E - \frac{1}{2} A \sigma^2 \\ &= r_f + \left(\frac{E[r_T] - r_f}{\sigma_T} \right) \sigma - \frac{1}{2} A \sigma^2 \end{aligned}$$

- Take the derivative with respect to σ and equate to zero:

$$\sigma^* = \frac{1}{A} \left(\frac{E[r_T] - r_f}{\sigma_T} \right)$$

- This is the optimal (target) amount of risk. Substitute back into E and we can find the corresponding expected return.

$$E^* = r_f + \frac{1}{A} \left(\frac{E[r_T] - r_f}{\sigma_T} \right)^2$$

- What is the position in the risky portfolio (the tangency portfolio) that yield this pair of expected return and risk?

Obtaining the Optimal Allocation - 3

- We know that for given x in Tangency portfolio and $(1-x)$ in risk free rate, the portfolio p on CAL must satisfy

$$E[r_p] = r_f + x(E[r_T] - r_f)$$

- Equating the left hand side with the optimal E^* we obtained earlier, we find

$$r_f + x(E[r_T] - r_f) = r_f + \frac{1}{A} \left(\frac{E[r_T] - r_f}{\sigma_T} \right)^2$$

- This gives the optimal position in the risky asset T:

$$x^* = \frac{1}{A} \frac{E[r_T] - r_f}{\sigma_T^2}$$

- **Example:** Let $E[r_T] - r_f = 6\%$, $\sigma_T = 15\%$, then
 - If $A = 2$, then $x^* = 133.3\%$
 - If $A = 4$, then $x^* = 66.6\%$
 - If $A = 8$, then $x^* = 33.3\%$